

STAKEHOLDER 22

Health and safety officers

80 decision-ready questions across eight operational categories

CATALOGUE DOMAINS

01	Policies, responsibilities and risk registers	HS-001 to HS-010
02	Hazards, inspections and workplace conditions	HS-011 to HS-020
03	Occupancy, activities and environmental evidence	HS-021 to HS-030
04	Incidents, near misses and corrective actions	HS-031 to HS-040
05	Contractors, permits and high-risk work	HS-041 to HS-050
06	Training, competence and communications	HS-051 to HS-060
07	Emergency preparedness and welfare arrangements	HS-061 to HS-070
08	Audits, trends, assurance boundaries and improvement planning	HS-071 to HS-080



EVIDENCE POSTURE

The May 2021 floor plan is historical spatial reference only. Current authoritative owners and competent professionals take precedence. Every sensor, integration and analytic is proposed until independently mapped, commissioned, permissioned and shown healthy. An explicit Unknown is safer than an unsupported safety, compliance, health or authorisation claim.

Purpose, scope and non-negotiable boundaries

A decision-support specification for a future service - not evidence that any deployment or integration currently exists.

PURPOSE

Frames practical questions health and safety officers may ask. Each card identifies the decision, proposed evidence, authoritative source, analytical operation, uncertainty, safe failure and concrete officer action.

EVIDENCE STATUS

Treat every sensor, telemetry route, integration and analytic as proposed until deployment, location mapping, commissioning, permission and current health are independently verified.

SPATIAL REFERENCE

The May 2021 floor plan supports historical geometry only. It cannot establish current room use, access, occupancy limit, services, hazards, closures, ownership or emergency arrangements.

AUTHORITATIVE PRECEDENCE

Current controlled records, named source owners, competent professionals and authorised incident or emergency command take precedence over telemetry and analytics. Conflicts remain visible until the correct owner resolves them.

PEOPLE AND PRIVACY

Do not identify or track people; infer health, disability, pregnancy, intent, blame, competence, productivity or individual risk; inspect research content, code, speech or network content; or create medical or individual risk records.

DECISION BOUNDARY

The service must not diagnose harm, certify safety or compliance, provide legal or occupational-health judgement, accept risk, issue permits, authorise work, command emergency action or control safety-critical assets.

How to read every question card

The six fields separate the operational decision, proposed evidence, authority, method, limits and action.

1 HEALTH AND SAFETY OFFICERS QUESTION

A natural operational question specific enough to expose the location, activity, decision interval and responsible owner, while allowing an honest Unknown.

2 SENSORS AND TELEMETRY

Physically plausible, privacy-preserving proposed evidence with modality, installation point, mapped asset or zone, coverage, interval, calibration or test state, clocks, uptime, gaps, drift and provenance.

3 AUTHORITATIVE SUPPORTING DATA

Current owners, controlled records and explicit precedence, with effective, observed and retrieved times, permissions, provenance, staleness and conflict handling.

4 ANALYSIS REQUIRED

The operation class - lookup, observation, calculation, estimate, forecast, comparison or recommendation - plus interval, minimum evidence, method, uncertainty and safe failure.

5 ANSWER BOUNDARY

What the service must not infer, disclose, promise, certify, authorise or control. The boundary protects people, legal roles, professional judgement and operational safety.

6 WHY THIS IS USEFUL

The concrete officer action: inspect, escalate, consult, brief, reschedule, restrict through existing authority, preserve evidence, repair a control or correct a source record.

Catalogue map: 80 questions across eight decision domains

Each category contains exactly ten questions and five two-card pages.

01	Policies, responsibilities and risk registers Confirm governing documents, accountable owners, review triggers, control dependencies and decision-ready escalation.	HS-001 to HS-010 10 questions 5 card pages
02	Hazards, inspections and workplace conditions Use reports, inspections and correctly scoped measurements to target competent checks and immediate controls.	HS-011 to HS-020 10 questions 5 card pages
03	Occupancy, activities and environmental evidence Relate anonymous space use and declared activities to current limits without tracking people or inferring health.	HS-021 to HS-030 10 questions 5 card pages
04	Incidents, near misses and corrective actions Support triage, evidence preservation, pattern analysis and action closure without blame or diagnosis.	HS-031 to HS-040 10 questions 5 card pages
05	Contractors, permits and high-risk work Check work interfaces, permits, isolations, readiness and handback without authorising work or controlling plant.	HS-041 to HS-050 10 questions 5 card pages
06	Training, competence and communications Find record, briefing and communication gaps while separating completion records from practical competence.	HS-051 to HS-060 10 questions 5 card pages
07	Emergency preparedness and welfare arrangements Test evidence for emergency systems, routes, role cover, welfare and exercises within current command arrangements.	HS-061 to HS-070 10 questions 5 card pages
08	Audits, trends, assurance boundaries and improvement planning Make assurance evidence reproducible, expose blind spots and prioritise proportionate improvement.	HS-071 to HS-080 10 questions 5 card pages

Policies, responsibilities and risk registers

HEALTH AND SAFETY OFFICERS QUESTION

HS-001 Which current corporate policy, local arrangement and risk-assessment set governs the proposed activity in this exact space and time window?

SENSORS AND TELEMETRY

No sensor can establish policy applicability. Proposed room, asset and service telemetry may only confirm named operational context after mapping, commissioning and access approval. **Timing:** use the requested decision window and the approved aggregation for the mapped zone or asset. **Evidence quality:** show exact coverage, clock status, latency, uptime, gaps and permission; never treat telemetry as a controlled document. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the corporate health and safety policy owner, local safety coordinator and activity owner; supporting sources: document control, space management and the current room owner. **Precedence:** the latest approved corporate policy and formally issued local arrangement outrank drafts, old copies and inferred practice. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the owner-defined review date or any superseding issue. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** on request and whenever activity, room or policy changes. **Minimum evidence:** a current location/activity identifier plus approved document versions. **Method:** document applicability and scope matching; match scope, exclusions, effective dates and linked assessments; route ambiguous applicability to the policy owner. **Uncertainty:** Document version lag, ambiguous scope, an imprecise activity description and superseded local copies can change the result. **Safe failure:** Return Applicability unresolved, display the competing documents and refer the question to the corporate policy owner.

ANSWER BOUNDARY

Do not interpret policy as legal advice, assume a historical floor-plan label proves current use, or present an expired or draft document as operative.

WHY THIS IS USEFUL

Confirm the governing controls and decision owner before commissioning an inspection, change, event response, investigation or permit review.

HEALTH AND SAFETY OFFICERS QUESTION

HS-002 Who is accountable for each high-priority risk during the decision window, and is a formally authorised deputy available if that owner cannot act?

SENSORS AND TELEMETRY

No sensor can determine accountability or competence. A proposed duty-status feed may show team-level coverage, not a person's whereabouts. **Timing:** use only approved rota windows; disclose no badge history. **Evidence quality:** report feed age, update cadence, permission scope and any unavailable or unmatched roles. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the risk-register owner and accountable local manager; supporting sources: the approved role directory, delegation register and duty rota owner. **Precedence:** the named approved risk owner and recorded delegation outrank directory guesses, email signatures or access events. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** one business day for duty coverage and the register review date for ownership. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** daily and before time-critical work or events. **Minimum evidence:** a live risk ID, current owner field and valid delegation window. **Method:** ownership join, delegation validation and gap detection; join risk, role and rota records without inferring presence; distinguish vacant, absent, unverified and permission-denied states. **Uncertainty:** A stale rota, a delegation not yet effective, a vacant role or an unmatched risk identifier can falsely imply cover. **Safe failure:** Return Owner or deputy not confirmed and escalate to the accountable manager before a time-critical decision.

ANSWER BOUNDARY

Do not infer availability, competence or accountability from badge use, calendar gaps or telemetry, and do not reassign a duty.

WHY THIS IS USEFUL

Route urgent risk decisions to the accountable manager, put authorised deputy cover in place and escalate ownerless high-priority risks.

Policies, responsibilities and risk registers

HEALTH AND SAFETY OFFICERS QUESTION

HS-003

Which risk assessments require competent review now because a material change, incident, complaint, inspection finding or verified control failure has occurred?

SENSORS AND TELEMETRY

Proposed triggers may include verified asset faults, service outages, changed room configuration, booking-type changes and incident timestamps. **Timing:** aggregate by assessment and day unless urgency requires events. **Evidence quality:** publish mapping, source health, clock alignment, false-trigger risk and missing integrations; a trigger is not proof of invalidity. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the risk-assessment register owner and activity owner; supporting sources: change control, incident management, Estates work orders and asset owners. **Precedence:** the approved assessment and formal review-trigger records outrank telemetry-derived cues. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the stated review date, or immediately when a verified owner records a material trigger. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and recommendation. **Decision interval:** daily, and within 15 minutes for verified safety-critical control loss. **Minimum evidence:** a current assessment linked to a due date or one verified trigger. **Method:** trigger-based review-queue generation; apply owner-approved trigger rules, deduplicate linked events and rank by consequence and evidence strength. **Uncertainty:** A trigger may be mislinked, duplicated, later withdrawn or timestamped differently from the assessment it appears to affect. **Safe failure:** Place the item in a human review queue as Trigger not confirmed; do not invalidate the assessment automatically.

ANSWER BOUNDARY

Do not declare an assessment invalid, determine legal sufficiency, invent a trigger from weak correlation or set a new review date for the owner.

WHY THIS IS USEFUL

Build a defensible reassessment queue, assign competent reviewers and act first on expired or materially changed high-consequence work.

HEALTH AND SAFETY OFFICERS QUESTION

HS-004

Which risk-register entries no longer match the current room use, equipment, occupancy assumptions, access arrangements or building services on which their controls depend?

SENSORS AND TELEMETRY

Proposed anonymous doorway counts, door and room-power states, and verified safety-service status may test only the operational assumptions named in a risk entry. Cover the exact mapped room, equipment and control dependency during the assessment window; use five- or fifteen-minute aggregates unless a fault event is required. Evidence envelope: show modality and installation point, mapping version, sample and aggregation interval, calibration or functional-test date, clock offset, uptime, gaps, drift or resets, permission and provenance. Do not use telemetry to classify room use. Unmapped or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the risk-register owner, room owner and space-management owner; supporting sources: timetabling, Security closure logs, Estates handback and event owners. **Precedence:** a current formal closure or competent owner instruction outranks bookings, historical plans and telemetry; approved room-use records outrank inferred use. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 15 minutes for closures, one day for bookings and the owner-defined register review date. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and comparison. **Decision interval:** before reliance on the assessment and after a material room, equipment, access or service change. **Minimum evidence:** current assessment version, named assumptions, current room/asset owner records and one verified observation or status source for each tested dependency. **Method:** compare each assumption with current controlled records and bounded observations; label untested assumptions separately. **Uncertainty:** Mapping age, unobserved assumptions, blind spots, aggregation loss and conflicting room or asset records can hide a mismatch. **Safe failure:** Return Assumption not verified and request current owner confirmation or competent inspection.

ANSWER BOUNDARY

Do not infer a specific activity or unauthorised behaviour from occupancy, override a closure, or treat the May 2021 plan as proof of current room function.

WHY THIS IS USEFUL

Identify assessments that need correction or reassessment before their controls are relied upon, and send each mismatch to the risk, room or asset owner.

Policies, responsibilities and risk registers

HEALTH AND SAFETY OFFICERS QUESTION

HS-005

Which recorded risk controls depend on a safety-critical asset or service that is unavailable, impaired, in test or not currently verified?

SENSORS AND TELEMETRY

Proposed status feeds may cover mapped alarms, ventilation, local exhaust, emergency lighting, access release, water and power. **Timing:** use change events and 5-minute health summaries. **Evidence quality:** require verified dependency mapping, commissioning, applicable calibration, clock alignment and heartbeat; distinguish fault, maintenance, disabled and unknown states, and show unmonitored dependencies. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the risk/control owner and relevant technical asset owner; supporting sources: BMS, fire-system records, CMMS, commissioning certificates and approved impairment logs. **Precedence:** the competent asset owner or official safety-system panel state outranks a generic dashboard or inferred fault. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 5 minutes for live impairments and the latest approved dependency map. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation and recommendation. **Decision interval:** within 5 to 15 minutes of a verified impairment. **Minimum evidence:** one approved control dependency plus a current authoritative asset state. **Method:** control-to-asset dependency graph evaluation and fault correlation; trace direct and upstream dependencies, avoid double counting and show alternative controls only when formally recorded. **Uncertainty:** Dependency maps may omit upstream services, a maintenance state may be ambiguous and a secondary dashboard may lag the official panel. **Safe failure:** Return Control availability unknown; do not rely on it until the asset owner confirms status and the risk owner decides any interim control.

ANSWER BOUNDARY

Do not automatically declare work safe or unsafe, reset or bypass equipment, invent substitute controls, or authorise continuation during an impairment.

WHY THIS IS USEFUL

Restrict, relocate or re-plan affected activity, raise the correct technical fault and track restoration of the depended-on control.

HEALTH AND SAFETY OFFICERS QUESTION

HS-006

Where is ownership of safety decisions unresolved at interfaces between the university, building owner, service teams, event organisers and contractors?

SENSORS AND TELEMETRY

No sensor can assign responsibility. Proposed zone-use counts and service-status events may show that an interface is active, but only as aggregated context. **Evidence quality:** report location mapping, time window, permissions and missing coverage; never use individual movement or device ownership as evidence of duty. **Timing:** use event windows for active interfaces and a daily responsibility-gap summary. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the local accountable manager and health and safety governance owner; supporting sources: leases, service agreements, room ownership, event bookings, contractor scopes and project responsibility matrices. **Precedence:** formally approved responsibility allocations and current contractual arrangements outrank custom, assumptions and historical ownership. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the latest signed issue or any recorded organisational change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at each change, event mobilisation and monthly governance review. **Minimum evidence:** a defined activity/service interface plus current responsibility records. **Method:** responsibility-interface and RACI gap analysis; map each decision, control, handoff and escalation to an owner; identify overlaps, gaps and permission-limited evidence. **Uncertainty:** Contracts, local arrangements and responsibility matrices may overlap, use different terms or lag organisational change. **Safe failure:** Return Interface owner unresolved and escalate to the local accountable manager before interface-dependent work proceeds.

ANSWER BOUNDARY

Do not assign legal duties, reinterpret contracts, disclose commercially restricted terms or infer responsibility from who happens to be present.

WHY THIS IS USEFUL

Resolve interface ownership before work begins, document escalation routes and stop hazards falling between teams, landlords, contractors or services.

Policies, responsibilities and risk registers

HEALTH AND SAFETY OFFICERS QUESTION

HS-007

What safety evidence is still missing before the proposed change can be presented to the responsible owner for a start, approval or commissioning decision?

SENSORS AND TELEMETRY

Proposed evidence may include asset-discovery changes, commissioning-test status, environmental baselines, door/route configuration and anonymous occupancy trials. **Timing:** use milestone snapshots, not continuous personal monitoring. **Evidence quality:** show mapping, test period, calibration, clock status, uptime and any untested modes. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the change owner, project manager and relevant safety/technical approvers; supporting sources: design records, commissioning results, risk assessments, room ownership and operational handback. **Precedence:** the approved change-control workflow and signed stage records outrank verbal agreement, telemetry or a planned completion date. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the current project stage and latest approved version. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** at each formal change-control gate and immediately before start, approval or commissioning. **Minimum evidence:** approved change scope, current hazard and risk record, consultation outcome, test plan, acceptance criteria, responsible owner and evidence for every safety dependency. **Method:** trace each gate criterion to a current record or verified test result; separate missing, failed, conditional and not-applicable items. **Uncertainty:** Untested interfaces, incomplete consultation, stale commissioning evidence and unresolved acceptance criteria can make a gate appear complete. **Safe failure:** Recommend Hold for evidence and give the responsible owner the exact missing tests, records or consultation; never approve the change.

ANSWER BOUNDARY

Do not approve the change, certify design or commissioning, accept residual risk, or treat a successful sensor reading as formal handback.

WHY THIS IS USEFUL

Give the responsible owner a precise hold list, assign missing tests or consultation, and prevent an unsupported change from reaching operation.

HEALTH AND SAFETY OFFICERS QUESTION

HS-008

Which risk-reduction actions are overdue, blocked, ownerless, duplicated or recorded complete without the verification required by the risk owner?

SENSORS AND TELEMETRY

Proposed work-order, asset-state and inspection-result telemetry may corroborate implementation. **Timing:** use daily status snapshots and event timestamps. **Evidence quality:** show asset mapping, completion evidence type, observation period, data latency and gaps; a closed work order or normal reading does not by itself prove control effectiveness. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the risk owner and corrective-action register owner; supporting sources: CMMS, project plans, procurement, budget owners and inspection records. **Precedence:** the risk owner's current action and residual-risk record outrank inferred priority; formal dependency and completion evidence outrank free-text status. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** one business day for action status and the next formal risk review for residual risk. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** weekly, at governance review and before any action is closed. **Minimum evidence:** current action ID, risk link, owner, due date, dependency, completion claim and owner-defined verification requirement. **Method:** classify overdue, blocked, ownerless, duplicate and unverified-complete actions; keep risk priority separate from action age. **Uncertainty:** Action ages can be distorted by revised due dates, duplicate records, unlinked dependencies and unsupported completion claims. **Safe failure:** Keep the action open or Unknown and route closure, duplication and priority decisions to the risk owner.

ANSWER BOUNDARY

Do not close actions, calculate legal tolerability, lower a risk rating automatically, assign blame or reprioritise commitments without the owner.

WHY THIS IS USEFUL

Recover stalled risk reduction, merge genuine duplicates and stop unsupported completion claims from hiding an active control gap.

Policies, responsibilities and risk registers

HEALTH AND SAFETY OFFICERS QUESTION

HS-009

Do current operational plans include the authorised accessibility and individual-adjustment dependencies needed for the activity, without exposing personal or medical information?

SENSORS AND TELEMETRY

No sensor may infer pregnancy, disability, health or assistance need. Proposed accessibility telemetry can cover only asset and route availability, such as lift status, powered-door status and refuge communication health, aggregated by zone. **Evidence quality:** show permission, mapping, uptime and gaps; no person-level linkage. **Timing:** use current asset state and authorised on-request checks aggregated by zone, with no person-level linkage. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the authorised adjustment-plan owner, occupational-health or HR service and emergency-planning owner; supporting sources: the individual, local manager, accessibility assessment and Estates asset owners. **Precedence:** the person-approved, need-to-know adjustment status and current emergency arrangement outrank generic assumptions; medical detail remains restricted. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the agreed review date or any person-approved change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** on authorised request and after relevant space or service change. **Minimum evidence:** a permissioned control requirement, current location and asset status without diagnosis data. **Method:** privacy-preserving control-completeness and asset-dependency check; match required arrangements to available routes, services and contacts; suppress small-group outputs and expose missing consent or access. **Uncertainty:** The authorised plan may be access-restricted, time-limited or dependent on an asset whose location or service state is not current. **Safe failure:** Return Arrangement not verified and route the gap privately to the authorised adjustment-plan owner without disclosing the reason.

ANSWER BOUNDARY

Do not infer a condition, create a medical or individual risk record, reveal the reason for an adjustment, or judge whether the person can undertake an activity.

WHY THIS IS USEFUL

Verify the enabling arrangement without exposing personal details, repair missing assets and consult through the authorised adjustment process.

HEALTH AND SAFETY OFFICERS QUESTION

HS-010

Which unresolved risks, weakly evidenced controls and time-critical owner decisions require formal escalation at the next safety governance meeting?

SENSORS AND TELEMETRY

Proposed dashboards may supply aggregated incidents, actions, asset impairments, inspection coverage, environmental exceptions and data-quality indicators. **Timing:** use weekly or monthly aggregates with suppressions for small groups. **Evidence quality:** state coverage, denominator, sampling, calibration, clock alignment, uptime and known blind spots. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the committee secretariat, health and safety governance owner and each contributing source owner; supporting sources: risk, incident, action, training, Estates and emergency-planning systems. **Precedence:** approved meeting scope and controlled source records outrank dashboard summaries; each source owner retains authority for interpretation. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the reporting cut-off, with late changes clearly timestamped. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation and recommendation. **Decision interval:** before each meeting and at the reporting cut-off. **Minimum evidence:** an approved agenda, reporting period, metric definitions and source-owner access. **Method:** evidence-pack assembly, completeness checking and provenance validation; compile only in-scope evidence, reconcile duplicates, label facts versus estimates and produce an exception list. **Uncertainty:** Reporting cut-offs, inconsistent risk scales, missing owners and selective source access can distort what appears most urgent. **Safe failure:** Issue an evidence-gap agenda rather than a recommendation and defer any unsupported decision to the accountable governance owner.

ANSWER BOUNDARY

Do not imply committee endorsement, hide adverse data, fill missing evidence with assumptions, or present an analytical summary as certification or legal assurance.

WHY THIS IS USEFUL

Run a focused governance meeting, separate decisions from information gaps and leave each action with an owner, deadline and evidence requirement.

Hazards, inspections and workplace conditions

HEALTH AND SAFETY OFFICERS QUESTION

HS-011

Which escape routes, final exits, fire doors or refuge approaches need immediate physical inspection because status is obstructed, secured incorrectly, impaired or unknown?

SENSORS AND TELEMETRY

Proposed door contacts, lock/release states and fixed time-of-flight or beam sensors at agreed choke points can flag change, not prove route clearance. **Timing:** use events plus 5-minute summaries during occupancy. **Evidence quality:** publish exact coverage, obstruction-detection limits, commissioning, clock alignment, battery/uptime and false-alarm history. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the fire-safety responsible person and current emergency-plan owner; supporting sources: the door register, Security closure log, Estates faults and competent inspection records. **Precedence:** a formal closure or competent fire-safety instruction outranks sensor state; current fire strategy and door designation outrank the historical floor plan. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 5 minutes for operational status and the latest approved fire-safety record. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation and recommendation. **Decision interval:** live to 5 minutes while the building is occupied. **Minimum evidence:** a mapped designated route or door plus one current authoritative state and usable telemetry where installed. **Method:** rule-based exception detection and source reconciliation; detect held-open, locked, unavailable and possible obstruction states; require human confirmation for ambiguous physical conditions. **Uncertainty:** Contacts and beams cannot see every obstruction, door-leaf defect or refuge condition; mapping, test mode and status latency may also mislead. **Safe failure:** Return Route not physically verified and request immediate competent inspection while retaining any existing restriction.

ANSWER BOUNDARY

A door contact, beam or counter must not be presented as proof that an escape route is physically clear. Do not certify an escape route, infer why a door is open, identify occupants, unlock or lock a door, order evacuation, or replace competent fire inspection.

WHY THIS IS USEFUL

Prioritise a physical route inspection, place an interim control through existing procedures and direct the fault to the correct owner.

HEALTH AND SAFETY OFFICERS QUESTION

HS-012

Which reported locations need prompt inspection for slip, trip, collision, falling-object or route-obstruction hazards, and what immediate control is already recorded?

SENSORS AND TELEMETRY

Proposed leak ropes, local floor-moisture sensors, door/route obstruction sensors and lighting status may cover known high-risk points. **Timing:** use event alerts and 5-minute persistence. **Evidence quality:** show installation footprint, surface sensitivity, cleaning interference, calibration, clock status, uptime and blind spots; most floor hazards still require inspection. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the local area manager and inspection/cleaning service owners; supporting sources: incident and near-miss reports, Estates faults, Security reports and active work zones. **Precedence:** a verified report or competent inspection outranks a sensor cue; formal closures and work-zone controls outrank normal route status. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 5 minutes for active hazards and one shift for routine reports. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation and recommendation. **Decision interval:** within 5 to 15 minutes for alerts and each inspection round. **Minimum evidence:** one verified high-confidence alert or two independent lower-confidence cues in the same zone. **Method:** conservative hazard-cue ranking; combine persistence, route criticality, occupancy opportunity and recent reports; suppress ranking where coverage is absent. **Uncertainty:** The reported location may be imprecise, the condition transient, the sensor cue false and the recorded temporary control stale. **Safe failure:** Return Hazard not confirmed, dispatch a site check and treat the immediate-control state as Unknown until observed.

ANSWER BOUNDARY

Do not treat a report or sensor cue as proof of a hazard, decide that a route is safe without inspection, or disclose a reporter unnecessarily. No safety or compliance certification, work authorisation or asset control.

WHY THIS IS USEFUL

Send an inspector to the right location, confirm the temporary control already in place, and escalate cleaning, cordoning, repair or storage action through the responsible owner.

Hazards, inspections and workplace conditions

HEALTH AND SAFETY OFFICERS QUESTION

HS-013

Which routes or task areas need a competent lighting assessment because failed luminaires or screening measurements may not support safe movement or work?

SENSORS AND TELEMETRY

Proposed calibrated lux sensors at representative route and task-plane locations, plus luminaire-driver or circuit status, may support screening. **Timing:** use 1-minute readings and 15-minute summaries for the mapped route or task area. **Evidence quality:** show sensor height and orientation, daylight influence, calibration date, clock alignment, coverage, uptime and outliers; one fixed sensor is not room-wide evidence. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the Estates lighting asset owner and local area/activity owner; supporting sources: lighting design information, inspection records, fault tickets and current room use. **Precedence:** competent design or inspection criteria for the actual task outrank generic thresholds and telemetry; verified circuit faults outrank inferred failure. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 15 minutes for readings and the latest approved task/use information. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation and comparison. **Decision interval:** every 15 minutes during relevant use and on fault events. **Minimum evidence:** a commissioned representative sensor or verified luminaire fault plus current task/route context. **Method:** threshold and persistence screening against locally approved criteria; compare time-matched values, account for daylight and sensor uncertainty, and identify areas requiring a competent lighting survey. **Uncertainty:** Sensor position may not represent the task plane, daylight and screening can change rapidly, and calibration or luminaire mapping may be wrong. **Safe failure:** Return Lighting adequacy unknown and arrange a competent survey or temporary owner-approved control without a compliance claim.

ANSWER BOUNDARY

A fixed lux reading must not be treated as a task-specific lighting assessment across an unmeasured area. Do not certify lighting compliance, infer personal visual ability, treat a sensor outside the task plane as exposure evidence, or switch circuits.

WHY THIS IS USEFUL

Target luminaire repair, temporary lighting, relocation or a formal illuminance survey at the affected route or task plane.

HEALTH AND SAFETY OFFICERS QUESTION

HS-014

Which occupied work zones need competent investigation because temperature, humidity, carbon-dioxide or ventilation-service evidence is outside the approved screening envelope?

SENSORS AND TELEMETRY

Proposed temperature, relative-humidity and CO2 sensors, plus supply-airflow or plant state where available, may support a workplace-condition screen. **Timing:** sample each minute and report 15-minute and hourly summaries for mapped occupied zones. **Evidence quality:** show placement, calibration, drift, clock alignment, coverage, uptime and sensor saturation; CO2 is a ventilation proxy, not exposure or health evidence. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the Estates HVAC owner and local activity/room owner; supporting sources: BMS, room-use records, approved workplace-condition criteria, complaints and maintenance faults. **Precedence:** a competent investigation and current operating instruction outrank dashboard thresholds; verified plant faults outrank inferred ventilation quality. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 15 minutes for conditions, 5 minutes for plant faults and the current activity window. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation and comparison. **Decision interval:** every 15 minutes while occupied, with immediate review for verified plant loss. **Minimum evidence:** usable environmental data for the occupied interval plus current room use and sensor-health evidence. **Method:** persistence, anomaly and occupancy-context screening; compare against owner-approved screening bands, detect sustained deviation and distinguish local sensor anomaly from zone-wide evidence. **Uncertainty:** Placement, occupancy context, outdoor conditions, HVAC response lag and the limited interpretation of carbon-dioxide readings affect the screen. **Safe failure:** Return Screening exception or coverage insufficient and request competent workplace-condition investigation without diagnosing exposure or harm.

ANSWER BOUNDARY

CO2 and environmental readings are screening evidence only and must not be used to diagnose symptoms, exposure or ventilation compliance. Do not diagnose heat stress, illness or poor health, certify ventilation or indoor-air safety, infer who was exposed, or command HVAC operation.

WHY THIS IS USEFUL

Select zones for competent workplace-condition investigation, check ventilation or controls and avoid broad claims from isolated measurements.

Hazards, inspections and workplace conditions

HEALTH AND SAFETY OFFICERS QUESTION

HS-015

Which areas need a competent occupational-noise survey because plant, alarms or temporary work may be producing elevated area sound?

SENSORS AND TELEMETRY

Proposed integrating sound-level meters can record privacy-preserving LAeq and peak summaries without storing audio or speech. **Timing:** use 1-minute values and 15-minute aggregates at defined locations. **Evidence quality:** show microphone class, placement, calibration before/after, clock alignment, wind/impact artefacts, coverage and uptime; fixed area sound is not personal dose. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the activity or equipment owner and occupational safety/noise adviser; supporting sources: work permits, equipment schedules, previous acoustic surveys, complaints and maintenance records. **Precedence:** a competent noise assessment and approved task controls outrank fixed-sensor screening; task timing records outrank inferred source attribution. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 15 minutes for screening and the current permit/activity window. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation, estimate and recommendation. **Decision interval:** every 15 minutes during relevant work and after complaints. **Minimum evidence:** a calibrated meter with usable data plus verified task or equipment timing. **Method:** area-noise event detection and source-time correlation; compare to locally approved screening triggers, separate background and transient events, and request personal or task-specific assessment where needed. **Uncertainty:** Meter position, reflections, short peaks, source attribution and calibration determine whether an area summary represents the task. **Safe failure:** Return Noise risk not established and commission a competent occupational-noise survey before selecting controls.

ANSWER BOUNDARY

Area sound summaries must not be converted into an individual noise dose, hearing-risk judgement or medical conclusion. Do not record speech, infer hearing damage or personal exposure, identify a speaker or worker, certify compliance, or shut down equipment.

WHY THIS IS USEFUL

Commission a competent noise survey at the right location and time, then plan engineering or administrative controls from valid measurements.

HEALTH AND SAFETY OFFICERS QUESTION

HS-016

Which storage, stacking, housekeeping, waste or aisle-clearance conditions require a site check, and which evidence is only a report rather than a verified observation?

SENSORS AND TELEMETRY

Proposed fixed distance sensors at selected aisles, shelf load cells on instrumented racks and door-state sensors may flag known constraints. **Timing:** use 5-minute summaries and change events. **Evidence quality:** show exact instrumented footprint, load-cell capacity/calibration, object-detection limitations, clock alignment, uptime and unmonitored shelves; visual inspection remains primary. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the store or area owner and local inspection owner; supporting sources: inventory records, fire/route plans, cleaning reports, work orders and competent rack inspections. **Precedence:** a competent physical inspection and approved storage limit outrank telemetry; formal route designation outranks observed convenience. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** one inspection shift for housekeeping and immediately for verified overload or blocked route. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation and recommendation. **Decision interval:** each inspection round and within 15 minutes of a high-confidence alert. **Minimum evidence:** a current zone/asset map plus a verified report or usable sensor cue. **Method:** inspection exception consolidation and constrained-zone screening; rank by route criticality, stored-material hazard, persistence and previous failures; show unmonitored areas explicitly. **Uncertainty:** Only instrumented aisles or racks are observed; actual load distribution, stacking stability and conditions behind stored items remain unseen. **Safe failure:** Return Storage condition unknown and require a physical check; do not infer rack capacity or structural adequacy.

ANSWER BOUNDARY

Instrumented aisles or racks must not be generalised to unmonitored storage areas. Do not certify stack or rack stability, identify who placed an item, infer material contents from weight, or direct untrained staff to move hazardous loads.

WHY THIS IS USEFUL

Restore housekeeping and access, isolate unsafe storage and obtain a competent rack or load check where required.

Hazards, inspections and workplace conditions

HEALTH AND SAFETY OFFICERS QUESTION

HS-017

Does a reported leak, water ingress, dampness or condensation issue require urgent slip control, electrical assessment, isolation or building inspection?

SENSORS AND TELEMETRY

Proposed leak ropes, point-moisture sensors, local relative humidity, water-meter anomaly detection and nearby electrical-panel alarms may support a site check. **Timing:** retain event times and 5-minute persistence for the mapped area. **Evidence quality:** show sensor location and height, sensitivity, calibration, clock alignment, uptime and dry-area blind spots; moisture does not identify contamination or energisation. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the Estates water/building fabric owner and local area manager; supporting sources: electrical responsible person, fault tickets, inspection reports, cleaning response and asset drawings. **Precedence:** a competent electrical or building inspection and formal isolation state outrank sensor inference; verified leak reports outrank meter anomaly alone. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 5 minutes for active leaks and the current inspection or isolation record. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation, comparison and recommendation. **Decision interval:** within 5 minutes for active leakage and every 15 minutes for damp trends. **Minimum evidence:** one verified leak signal or two corroborating moisture/flow cues plus a mapped zone. **Method:** multi-sensor event correlation and area triage; classify likely active leak, condensation or unknown; check proximity to electrical assets and occupied routes without declaring cause. **Uncertainty:** The water source, hidden extent, intermittency and proximity to live electrical parts may be unknown, while a meter anomaly is non-specific. **Safe failure:** Treat the report as unverified but potentially urgent and request cordon and competent inspection through current owners; do not isolate remotely.

ANSWER BOUNDARY

Moisture telemetry must not be used to identify contamination, electrical energisation or structural cause. Do not diagnose mould, contamination or electrical safety, infer source responsibility, operate valves or isolators, or state that an area is safe to enter.

WHY THIS IS USEFUL

Cordon and inspect the area, call Estates or electrical response, isolate affected services and prevent secondary damage.

HEALTH AND SAFETY OFFICERS QUESTION

HS-018

Which guards, covers, interlocks, barriers or emergency stops are reported missing, defeated, damaged or unverified, and what asset must be checked before use?

SENSORS AND TELEMETRY

Proposed safety-relay, interlock, emergency-stop circuit and barrier-contact status may be read only from approved machinery interfaces. **Timing:** use event state and heartbeat, not continuous operator monitoring. **Evidence quality:** show asset mapping, fail-safe logic, diagnostic coverage, clock alignment and maintenance mode; telemetry cannot prove a physical guard is fitted correctly. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the machinery or technical asset owner and local activity supervisor; supporting sources: maintenance records, pre-use checks, inspection certificates, risk assessments and fault reports. **Precedence:** a competent physical inspection and approved machine status outrank a remote signal; formal out-of-service tagging outranks normal control-system state. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediate refresh for reported defects and 1 minute for approved safety-circuit status. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and recommendation. **Decision interval:** live to 1 minute during availability and before use. **Minimum evidence:** a mapped asset plus a verified report or trustworthy safety-circuit state. **Method:** safety-device state reconciliation and defect escalation; compare expected guard/interlock state with maintenance and isolation records; treat diagnostic ambiguity as unavailable. **Uncertainty:** A safety-circuit signal cannot prove guard integrity, correct asset identity, absence of bypass or safe reset after maintenance. **Safe failure:** Return Protective function not verified, keep the asset unavailable and require competent pre-use inspection.

ANSWER BOUNDARY

A healthy circuit state must not be treated as proof that a guard, barrier or emergency stop is physically correct and suitable. Do not declare machinery safe to use, reset, bypass or test a safety circuit remotely, infer operator behaviour, or authorise restart.

WHY THIS IS USEFUL

Take the named asset out of service, arrange competent inspection and prevent restart until guards and protective functions are verified.

Hazards, inspections and workplace conditions

HEALTH AND SAFETY OFFICERS QUESTION

HS-019

Are local exhaust ventilation, eyewash, safety shower or other emergency protection provisions available for the declared work, and what still requires a pre-use check?

SENSORS AND TELEMETRY

Proposed differential-pressure and airflow sensors, fan status, cabinet/valve contacts and instrumented flow-test results may cover selected provisions. **Timing:** use 1-minute status and test-event records. **Evidence quality:** show sensor location, calibration, test method, clock alignment, uptime and which assets still require manual checks; presence is not performance certification. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the laboratory/workshop manager and relevant technical asset owner; supporting sources: risk assessments, planned work, examination/test records, asset register and maintenance faults. **Precedence:** current competent examination evidence and formal asset availability outrank telemetry; the approved task assessment defines the required provision. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before each task and within 5 minutes of verified asset impairment. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and comparison. **Decision interval:** before work, daily during campaigns and on faults. **Minimum evidence:** a planned task, approved required controls and current asset/test status. **Method:** task-to-control dependency and readiness checking; match each control to location and availability, distinguish due, failed, isolated and unverified, and require physical checks where the asset is not instrumented. **Uncertainty:** Pressure or fan state may not prove capture velocity, eyewash performance, water quality or suitability for the declared substance and task. **Safe failure:** Return Protection provision not verified and delay or relocate the work until the owner completes the required pre-use or competent test.

ANSWER BOUNDARY

Availability or a recent test result must not be presented as certification that protection is suitable for the planned substance or task. Do not certify extraction performance or emergency equipment readiness, authorise hazardous work, or treat a normal status bit as proof of a successful flow test.

WHY THIS IS USEFUL

Delay or relocate the declared work, test the provision and obtain an owner-approved alternative before relying on it.

HEALTH AND SAFETY OFFICERS QUESTION

HS-020

Which inspection zones should be visited first because checks are overdue, repeatedly adverse, affected by change or unsupported by current traceable evidence?

SENSORS AND TELEMETRY

Proposed inspection check-in timestamps, non-personal zone scans and telemetry-health records can show coverage, not inspector performance. **Timing:** aggregate by zone and period. **Evidence quality:** show schedule mapping, missed uploads, clock alignment, duplicate scans, device uptime and uninstrumented hazards; do not track routes or time-on-task. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the inspection programme owner and each area owner; supporting sources: completed checklists, corrective actions, incident records, CMMS and sensor commissioning/health logs. **Precedence:** a signed competent inspection record outranks a scan or dashboard status; current zone ownership and inspection frequency outrank the historical plan. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the defined inspection interval and one day for record synchronisation. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: calculation, comparison and recommendation. **Decision interval:** daily for overdue zones and monthly for trend review. **Minimum evidence:** a current inspection schedule, zone map and completion or exception records. **Method:** coverage, recurrence and evidence-quality analysis; calculate due/complete/failed/unknown states, cluster repeated findings and separate no finding from no inspection. **Uncertainty:** Priority is sensitive to missing reports, different inspection frequencies, changing use and unequal sensor coverage between zones. **Safe failure:** If the ranking is unstable, use current due dates and consequence rules and ask area owners to set the inspection order.

ANSWER BOUNDARY

Do not infer negligence, productivity or inspector location, certify an entire zone from a sample, or close repeated findings automatically.

WHY THIS IS USEFUL

Deploy inspection capacity first to high-consequence locations with overdue checks, repeat findings or weak evidence coverage.

Occupancy, activities and environmental evidence

HEALTH AND SAFETY OFFICERS QUESTION

HS-021

Will expected occupancy remain within the current owner-approved room limit and the egress assumptions for the whole activity window?

SENSORS AND TELEMETRY

Proposed anonymous doorway or overhead time-of-flight counters can estimate entries, exits and zone totals without identity. **Timing:** use 30-second events, 5-minute peaks and booking-level summaries. **Evidence quality:** show counter placement, direction errors, re-entry uncertainty, calibration, clock alignment, uptime and coverage; counts do not establish room layout or behaviour. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the room/space owner and relevant fire-safety or activity owner; supporting sources: booking systems, event plans, temporary restrictions and approved layout records. **Precedence:** the current competent owner's operational limit and formal temporary restriction outrank booking capacity, historical plans and sensor estimates. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 5 minutes for live estimates and the current event/booking version. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: calculation, estimate and forecast. **Decision interval:** at planning, after layout or timetable change, and during the declared activity window where a verified live count is available. **Minimum evidence:** current owner-approved room limit and egress assumptions, declared layout and schedule, and either a bounded attendance estimate or healthy mapped anonymous counters. **Method:** calculate expected and observed occupancy against the approved limit with counter error bounds; test the entire window rather than a peak snapshot. **Uncertainty:** Counter error, missed entrances, tailgating, late arrivals, layout change and forecast range can affect the whole-window occupancy estimate. **Safe failure:** Return Occupancy not established and require owner-controlled counting or a more conservative attendance plan.

ANSWER BOUNDARY

An anonymous count is an estimate with reconciliation error; it must not identify people or prove the declared layout or activity. Do not set capacity, certify crowd safety, identify or track occupants, infer behaviour, or automatically lock, open or close a room.

WHY THIS IS USEFUL

Reduce attendance, change the layout or timetable, add authorised counting, or relocate the activity before approved capacity and egress assumptions are exceeded.

HEALTH AND SAFETY OFFICERS QUESTION

HS-022

Which restricted, closed or uncommissioned zones have conflicting booking, access, closure or occupancy evidence that requires owner confirmation?

SENSORS AND TELEMETRY

Proposed anonymous zone counts, door state, aggregate access transactions and declared equipment or service states may reveal a source conflict, not unauthorised use. **Timing:** compare 5-minute zone summaries only within the recorded restriction or closure window. **Evidence quality:** show count error, tailgating and pass-back limitations, adjacent-zone leakage, mapping, commissioning, clock alignment, uptime and permission; do not identify occupants. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the room owner, Security closure owner and commissioning or handback owner; supporting sources: current restrictions, access profiles, bookings, work orders and approved activity declarations. **Precedence:** an authorised closure, restriction or competent owner instruction outranks bookings, access transactions and telemetry; access permission is not proof of presence or authorised activity. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** live for active closure changes and 5 minutes for source health. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: comparison and recommendation. **Decision interval:** during each restriction or closure window. **Minimum evidence:** a current mapped zone, authoritative restriction and one independent permissioned operational source. **Method:** restricted-zone source-conflict detection; compare records and anonymous states, distinguish exact, partial, adjacent and unmapped matches, and route discrepancies to the zone owner. **Uncertainty:** Bookings, access transactions, counters and closure records describe different things and may have different effective times. **Safe failure:** Maintain the restriction, return Zone status unresolved and ask the current room or closure owner to confirm the authoritative state.

ANSWER BOUNDARY

Do not identify a person, infer trespass or activity, accuse misuse, override a closure or treat access permission, a door event or an anonymous count as proof of occupancy. The service must not infer health or individual risk, disclose restricted identities or medical information, diagnose harm, promise safety, certify safety or compliance, authorise work or control safety-critical assets.

WHY THIS IS USEFUL

Ask the current zone owner to confirm status, correct an access or booking conflict, inspect the area and preserve a restriction until the discrepancy is resolved.

Occupancy, activities and environmental evidence

HEALTH AND SAFETY OFFICERS QUESTION

HS-023

For each approved lone-working session, is the required check-in current and is the authorised welfare-response route available for the whole session?

SENSORS AND TELEMETRY

No occupancy sensor should establish lone-working status. A proposed check-in service may expose only session reference, mapped zone, due, received or overdue state and responder-channel health; anonymous counts may corroborate zone use but not identity. **Timing:** evaluate each agreed deadline and 5-minute service-health interval. **Evidence quality:** show clock drift, count error, adjacent-zone leakage, missing integrations, uptime and permission scope. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the lone-working policy owner and local activity manager; supporting sources: the approved session register, check-in log, duty-responder rota, room booking and Security operating record. **Precedence:** an explicit approved session and its check-in state outrank anonymous counts or access events; the authorised response plan governs escalation. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** each check-in deadline, any rota change and 5 minutes for channel health. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and recommendation. **Decision interval:** at every agreed check-in deadline and when the response channel changes. **Minimum evidence:** an active approved session, due time, mapped zone and authorised responder route. **Method:** session-deadline and response-route validation; compare due and received states, test channel availability and escalate only through the recorded welfare process; occupancy is context, not proof. **Uncertainty:** A missed check-in can reflect a service failure or schedule change, and the system cannot prove presence, welfare or responder availability. **Safe failure:** Invoke the authorised welfare-response process as Session status unknown; do not start movement tracking.

ANSWER BOUNDARY

Do not infer that a named person is present, absent, safe or in distress from check-in or occupancy evidence. Do not create lone-working status from occupancy, identify or track a worker, infer distress, reveal personal contact details or contact anyone outside the approved response plan. The service must not infer health or individual risk, disclose medical information, diagnose harm, promise safety, certify safety or compliance, authorise work or control safety-critical assets.

WHY THIS IS USEFUL

Trigger the authorised check-in or welfare response, repair a failed contact route and improve cover without creating movement surveillance.

HEALTH AND SAFETY OFFICERS QUESTION

HS-024

Where do anonymous flow measurements indicate crowding, long queues or opposing pedestrian movements that merit a route-management check?

SENSORS AND TELEMETRY

Proposed overhead time-of-flight or low-resolution thermal counters can provide anonymous zone density and directional flow at agreed pinch points. **Timing:** use 30-second samples and 5-minute peaks. **Evidence quality:** show field of view, occlusion, group-count error, calibration, clock alignment, uptime and unmonitored routes; store no images or identities. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the event/space owner and Security operational lead; supporting sources: event schedule, route plan, temporary closures, steward observations and fire-safety arrangements. **Precedence:** a competent event or Security instruction and formal route closure outrank sensor estimates; approved route capacity outranks historical geometry alone. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 1 to 5 minutes during events and the current route-plan version. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation, estimate and recommendation. **Decision interval:** every minute during planned peaks. **Minimum evidence:** a mapped pinch point, approved route context and usable anonymous counts. **Method:** density, queue and directional-flow screening; calculate occupancy bands, queue persistence and opposing-flow indicators with uncertainty; require observation when occlusion or coverage is poor. **Uncertainty:** Occlusion, group-count error, unmonitored entrances, temporary route changes and an outdated crowd threshold can alter the result. **Safe failure:** Return Crowding not reliably established and request steward observation plus conservative route management.

ANSWER BOUNDARY

Do not reconstruct paths, identify people or treat a density estimate as proof of unsafe behaviour. Do not identify attendees, infer emotion or intent, certify crowd safety, issue emergency commands, or directly control gates and doors.

WHY THIS IS USEFUL

Change queue layout, stewarding, routing or timing at the evidence-supported pinch points and then recheck conditions.

Occupancy, activities and environmental evidence

HEALTH AND SAFETY OFFICERS QUESTION

HS-025

Which teaching, research, maintenance, cleaning, delivery or event activities overlap in time and place with an unmanaged safety interface?

SENSORS AND TELEMETRY

Proposed room-state, anonymous occupancy, loading-area, lift and safety-service telemetry can corroborate whether planned interfaces are active. **Timing:** use event timestamps and 15-minute summaries. **Evidence quality:** show location mapping, activity ambiguity, clock alignment, uptime and missing systems; telemetry must not classify research or teaching content. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the room/event owner and permit or work-order owner; supporting sources: timetabling, delivery bookings, cleaning plans, Security closures and Estates schedules. **Precedence:** a formal permit, closure or competent owner instruction outranks calendar entries and telemetry; current approved timings outrank cached schedules. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 15 minutes for active work and one day for future planning. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** at booking, daily look-ahead and every 15 minutes during critical interfaces. **Minimum evidence:** two current activities with overlapping time/location plus applicable separation rules. **Method:** spatial-temporal schedule conflict detection; compare work zones, routes, noise/dust/services and required controls; label potential conflicts rather than assumed activity. **Uncertainty:** Activities may be undeclared, delayed or described at different spatial granularity, and the hazardous interface may be outside sensor coverage. **Safe failure:** Return Interface cannot be ruled out and require affected owners to separate, reschedule or explicitly control the overlap.

ANSWER BOUNDARY

Do not infer actual activity from sensors, inspect research content, cancel bookings, authorise co-activity or reveal confidential project details.

WHY THIS IS USEFUL

Reschedule, segregate or coordinate overlapping activities and document the owner for each shared interface before work starts.

HEALTH AND SAFETY OFFICERS QUESTION

HS-026

For the declared task, did the measured environmental conditions remain within its approved operating envelope for the full activity window, and where is coverage insufficient?

SENSORS AND TELEMETRY

Proposed temperature, humidity, CO2, pressure, noise and lighting evidence may be used only where the activity owner has defined relevant metrics. **Timing:** sample at 1 minute and aggregate to 5 or 15 minutes. **Evidence quality:** show placement, calibration, clock alignment, occupancy coverage, uptime, censoring and gaps; no personal exposure measurement is implied. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the activity owner and relevant technical/environmental owner; supporting sources: approved operating limits, booking records, BMS, equipment logs and fault records. **Precedence:** the activity-specific approved envelope and competent measurement outrank generic comfort bands; verified fault and calibration records qualify telemetry. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the exact activity interval plus the latest approved limits. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation, calculation and comparison. **Decision interval:** after each relevant activity and during sustained excursions. **Minimum evidence:** time-aligned activity start/end, approved limits and usable representative measurements. **Method:** retrospective interval comparison and exception-duration calculation; calculate duration, magnitude and coverage of excursions, account for sensor uncertainty and distinguish unknown intervals from compliant intervals. **Uncertainty:** Task-envelope applicability, sensor placement, calibration, data gaps and time alignment can prevent a full-window comparison. **Safe failure:** Return Operating-envelope conformance not demonstrated and refer the start or continuation decision to the task owner and competent assessor.

ANSWER BOUNDARY

Point measurements must not be generalised beyond mapped representativeness or converted into personal exposure or health conclusions. Do not certify safe exposure, infer who was present or affected, diagnose harm, or declare the activity compliant from partial or proxy evidence.

WHY THIS IS USEFUL

Pause or adjust the activity, repeat measurement, correct the environment and revise the operating envelope or monitoring plan.

Occupancy, activities and environmental evidence

HEALTH AND SAFETY OFFICERS QUESTION

HS-027

Have temporary room, furniture, route or access changes invalidated any evacuation-assistance or accessible-egress arrangement?

SENSORS AND TELEMETRY

Proposed powered-door, lift, refuge-call and route-state telemetry can show asset availability; anonymous counts may show use pressure. **Timing:** use 5-minute status and change events. **Evidence quality:** show exact route coverage, commissioning, clock alignment, uptime and unmonitored obstacles. No sensor may identify a person or infer an assistance need. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the emergency-planning and accessibility-arrangement owners; supporting sources: room layout/change control, Estates asset status, Security closures and person-approved adjustment plans. **Precedence:** current person-approved arrangements and competent emergency/accessibility assessment outrank generic assumptions; formal closures outrank planned layouts. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediately after layout/access change and within 5 minutes for critical asset loss. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** at each room change, booking setup and service impairment. **Minimum evidence:** a current layout/restriction plus permissioned assistance requirements expressed without medical detail. **Method:** change-impact and dependency matching; compare routes, refuges, lifts, doors and support roles; suppress individual details and require consultation for ambiguous effects. **Uncertainty:** Temporary furniture or access changes may be undocumented, route assets may be impaired and personal arrangements may be permission-restricted. **Safe failure:** Return Accessible-egress arrangement not verified and consult the authorised emergency and accessibility owners before occupation.

ANSWER BOUNDARY

Do not infer disability, disclose personal emergency plans, decide whether evacuation assistance is adequate, or direct an individual without the authorised plan.

WHY THIS IS USEFUL

Update route plans, room setup, stewarding and person-agreed arrangements before the changed space is used.

HEALTH AND SAFETY OFFICERS QUESTION

HS-028

Is the declared activity compatible with the room's current authorised safety profile, restrictions, installed controls and supervision requirements?

SENSORS AND TELEMETRY

Proposed anonymous occupancy and explicitly declared equipment or service states may show that a booked zone is active; they must not classify the activity. **Timing:** use 5-minute zone and mapped-asset summaries only within the declared activity window. **Evidence quality:** show mapping confidence, commissioning, clock alignment, uptime, alternative explanations and missing task declarations; do not inspect network traffic, code, speech or research content. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the room owner and activity owner; supporting sources: the approved room-use profile, booking or permit, task risk assessment, change control, equipment register and current restrictions. **Precedence:** the authorised activity declaration and competent room-owner decision outrank analytical cues or historical labels; formal closures and restrictions govern. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the current booking or permit window and every room-use, closure or task change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** before start and when the activity, layout or restriction changes. **Minimum evidence:** a current declared activity category, mapped zone, approved space profile and linked controls. **Method:** declared-activity-to-space-profile compatibility checking; compare explicit requirements and prohibitions; use occupancy or asset state only to flag that owner confirmation may be urgent, never to infer content. **Uncertainty:** A vague activity description, outdated room profile, missing supervision detail or unverified installed control can create a false match. **Safe failure:** Return Activity compatibility unresolved and hold the start for room-owner and activity-owner review.

ANSWER BOUNDARY

Do not infer an activity from environmental, power, network or occupancy patterns, inspect research or teaching content, accuse unauthorised use, override a closure or authorise continuation.

WHY THIS IS USEFUL

Ask the room and activity owners to confirm the use, add the required controls or move the work before an unsupported activity continues.

Occupancy, activities and environmental evidence

HEALTH AND SAFETY OFFICERS QUESTION

HS-029

Which occupied zones coincide with a verified impairment of ventilation, alarm, access release or another required safety-critical service?

SENSORS AND TELEMETRY

Proposed anonymous zone counts may be combined with mapped ventilation, alarm, emergency-lighting or access-release impairment states only after commissioning and permission approval. **Timing:** use 1-minute service events and 5-minute zone-presence summaries, suppressing small groups. **Evidence quality:** show count uncertainty, zone leakage, service-map version, clock alignment, latency, uptime and blind spots; absence of a count is not proof that a zone is empty. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the safety-service owner, incident or impairment owner and room owner; supporting sources: approved occupancy limits, closures, bookings, local contingency arrangements and authorised manual checks. **Precedence:** a formal closure, evacuation or competent owner direction outranks occupancy analytics; the service owner determines impairment status. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** live events and 5 minutes for occupancy context. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation, comparison and recommendation. **Decision interval:** while an impairment is active. **Minimum evidence:** a mapped impairment with affected zones plus privacy-approved anonymous occupancy evidence or an authorised human report. **Method:** occupied-zone and verified-service-impairment intersection; intersect zone scope and time, apply small-group suppression, show uncertainty and list unaffected or unmonitored areas. **Uncertainty:** Occupancy estimates, impairment-zone mapping and event clocks may not align, and an upstream fault may affect more or fewer spaces than recorded. **Safe failure:** Return Co-occurrence not verified, escalate the impairment and keep conservative restrictions until the competent owner confirms scope.

ANSWER BOUNDARY

Do not identify or locate individuals, infer exposure or harm, promise that a zone is empty, issue an evacuation instruction or override incident command. The service must not disclose restricted identities or medical information, diagnose harm, promise safety, certify safety or compliance, authorise work or control safety-critical assets.

WHY THIS IS USEFUL

Prioritise the authorised zone check, restrict or relocate activity, communicate the impairment and repair the service without relying on person tracking.

HEALTH AND SAFETY OFFICERS QUESTION

HS-030

What minimum evidence is still missing before the responsible person can make the planned start-or-continue decision?

SENSORS AND TELEMETRY

Proposed evidence may combine only the task-defined environmental metrics, mapped safety-service states and source-health indicators required by the approved activity plan. **Timing:** use the stated pre-start window, live fault events and 5- or 15-minute summaries defined by the responsible owner. **Evidence quality:** show coverage, calibration, mapping, clock offset, latency, uptime, gaps and permission denial; absence of an alert is not evidence of readiness. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the activity owner, room owner and relevant technical asset owners; supporting sources: the approved risk assessment, permit or booking, operating envelope, inspection evidence, impairment log and current restrictions. **Precedence:** the responsible person's decision and current controlled records outrank telemetry or a historical floor-plan label. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediately before start, at every material change and within the owner-defined limit for live impairments. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** on request, immediately before start and whenever conditions change. **Minimum evidence:** the owner-approved checklist for location, activity, people-independent controls, service health and environmental limits. **Method:** pre-start or continue-readiness evidence gate; test each item for presence, currency, scope and threshold; list missing, stale, conflicting and inspection-only items without producing a go or no-go authorisation. **Uncertainty:** The task declaration, permit, inspection, supervision or control dependency may be missing, stale or owned in a restricted system. **Safe failure:** Return Start or continue evidence incomplete with a precise missing-evidence list; never convert completeness into approval.

ANSWER BOUNDARY

Do not make the start or continue decision, treat a passing sensor value as proof that the activity is safe, fill missing evidence from another room or use the May 2021 plan as current operational truth.

WHY THIS IS USEFUL

Give the responsible person a precise pre-start evidence gap list rather than a false ready or not-ready verdict.

Incidents, near misses and corrective actions

HEALTH AND SAFETY OFFICERS QUESTION

HS-031 Which newly reported incidents or near misses meet current criteria for immediate human triage, emergency escalation or evidence preservation?

SENSORS AND TELEMETRY

Proposed verified alarm, leak, asset-fault, route-blockage and anonymous occupancy events may add operational context to a submitted report. **Timing:** use event timestamps and 5-minute summaries. **Evidence quality:** show mapping, clock alignment, alert quality, uptime and missing feeds; telemetry cannot diagnose injury or replace the reporter's account. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the incident-management owner and duty health and safety lead; supporting sources: Security, first-aid, Estates and local manager records. **Precedence:** the current escalation matrix and competent duty decision outrank automated severity; emergency-service or official incident status outranks local telemetry. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediate receipt and refresh within 5 minutes for new reports. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** on submission and on any verified escalation event. **Minimum evidence:** a report with time/location and at least one factual severity or control indicator. **Method:** rule-based triage cueing and missing-information detection; apply approved escalation rules conservatively, separate reported facts from corroboration and route ambiguous cases upward. **Uncertainty:** Initial reports may be incomplete or duplicated, severity may be unknown and escalation criteria or emergency status may change quickly. **Safe failure:** When a time-critical criterion cannot be excluded, route immediate human triage and preserve the original evidence.

ANSWER BOUNDARY

Do not diagnose injury, determine statutory reportability, score an individual's risk, minimise a report, identify witnesses unnecessarily or dispatch emergency services autonomously.

WHY THIS IS USEFUL

Route the case to the correct responder, secure the scene, preserve volatile evidence and request the few facts needed for immediate triage.

HEALTH AND SAFETY OFFICERS QUESTION

HS-032 What is the most defensible event timeline from permissioned records and observations, with clock quality and uncertainty shown?

SENSORS AND TELEMETRY

Proposed evidence may include incident timestamps, mapped alarm and asset events, environment readings, access-zone aggregates and explicitly submitted photographs or observations. **Timing:** retain source-native event times and create a minute-level sequence only where clock quality supports it. **Evidence quality:** show clock offsets, latency, missing intervals, duplicate events, zone uncertainty, redaction status and source health; do not reconstruct individual movements. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the incident-management owner and each contributing system owner; supporting sources: authorised witness accounts, work orders, permits, maintenance logs, communications records and evidence-preservation decisions. **Precedence:** authenticated original records and competent investigation decisions outrank derived ordering or telemetry correlation. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** preserve immediately and refresh as authenticated evidence arrives; expose conflicts rather than forcing a single sequence. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and comparison. **Decision interval:** at initial triage and each evidence update. **Minimum evidence:** the reported event window, source clocks and at least one authenticated event or observation. **Method:** multi-source temporal reconciliation; normalise times, retain ranges where clocks disagree, distinguish observed, reported and inferred links, and minimise identity fields. **Uncertainty:** Clock skew, time-zone conversion, delayed manual entry, missing intervals and uncertain event identity can change apparent sequence. **Safe failure:** Publish a bounded timeline with Unknown intervals and do not invent order, causality or a single definitive time.

ANSWER BOUNDARY

Do not fill timestamp gaps by inventing event order or reconstructing individual movements. Do not infer identity, intent, blame, causation, injury or an individual route from correlated events, and do not disclose witness or affected-person details beyond authorised roles. The service must not diagnose harm, promise safety, certify safety or compliance, authorise work or control safety-critical assets.

WHY THIS IS USEFUL

Preserve a traceable timeline for investigator use, identify clock or coverage gaps and avoid unsupported reconstruction.

Incidents, near misses and corrective actions

HEALTH AND SAFETY OFFICERS QUESTION

HS-033

Which incidents or near misses show a credible recurring pattern by hazard, asset, zone, activity condition or failed control?

SENSORS AND TELEMETRY

Proposed asset state and environmental summaries can be linked only at asset/zone and period level. **Timing:** use daily or event-window aggregates with minimum-count suppression. **Evidence quality:** show taxonomy quality, location mapping, sampling, calibration, clock alignment, uptime and changed coverage; no individual or team performance analysis. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the incident-data owner and risk owner; supporting sources: asset register, inspection findings, work orders, bookings and local hazard classifications. **Precedence:** validated incident coding and current asset/location identifiers outrank free-text similarity; competent investigation findings outrank statistical association. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** monthly for trend review and after a serious recurrence. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: calculation, estimate and comparison. **Decision interval:** monthly, quarterly and on a verified repeat event. **Minimum evidence:** at least three comparable events or an owner-defined serious repeat, with stable coding and denominator where available. **Method:** cluster, recurrence-rate and context-association analysis; group by hazard/asset/location, normalise by activity opportunity, test sensitivity to coding gaps and report confidence intervals. **Uncertainty:** Reporting practice, exposure opportunity, duplicate events, classification drift, seasonality and small counts can imitate recurrence. **Safe failure:** Return Recurrence not established and flag the pattern for competent investigation rather than root-cause or blame attribution.

ANSWER BOUNDARY

A recurring association must not be presented as proof of cause, blame or deficient individual performance. Do not infer causation, blame a person or team, create individual risk scores, disclose small-group patterns or claim that non-reporting means no incidents.

WHY THIS IS USEFUL

Open a focused investigation, inspect recurring assets or zones and redesign the control where the pattern is credible.

HEALTH AND SAFETY OFFICERS QUESTION

HS-034

Which incident corrective actions are overdue, ownerless, blocked by dependency or recorded complete without the agreed verification?

SENSORS AND TELEMETRY

Proposed work-order and asset-state events may corroborate progress, while document workflows supply timestamps. **Timing:** use daily snapshots. **Evidence quality:** show linkage confidence, completion evidence, clock status, system latency, duplicate tasks and unmapped actions; normal telemetry after repair is supportive, not closure evidence by itself. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the incident/action owner and accountable risk owner; supporting sources: CMMS, procurement, project plans, contractor records and local inspection evidence. **Precedence:** the approved corrective-action register and owner sign-off outrank inferred completion; verified dependencies and formal handback outrank free-text progress. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** one business day for status and the agreed due date. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: calculation, comparison and recommendation. **Decision interval:** daily for overdue items and weekly for escalation review. **Minimum evidence:** an action ID, due date, current owner, linked incident/risk and status evidence. **Method:** action ageing, ownership and dependency-network analysis; calculate overdue age, blocker chain and unowned states; distinguish implemented, verified, ineffective, superseded and administratively closed. **Uncertainty:** Owner and due-date fields may lag, dependencies may be unlinked, verification criteria absent and completion disputed. **Safe failure:** Keep the corrective action open or Unknown and send the disputed state to the action and risk owners.

ANSWER BOUNDARY

Do not close an action, reassign accountability, infer negligence, reduce risk automatically or treat a purchase order as completed control.

WHY THIS IS USEFUL

Escalate the actual blocker, assign an owner and resources, and keep high-consequence actions visible until independently verified.

Incidents, near misses and corrective actions

HEALTH AND SAFETY OFFICERS QUESTION

HS-035

After a corrective control is implemented, is there credible case-level evidence that the intended indicator changed without creating an obvious new hazard elsewhere?

SENSORS AND TELEMETRY

Proposed before/after evidence may include relevant asset faults, environmental metrics, route counts, alarm frequency and incident rates at matched aggregation. **Timing:** keep sampling and sensor placement consistent. **Evidence quality:** show baseline length, calibration, clock alignment, uptime, confounders and changed coverage; use no personal productivity or health measures. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the corrective-action owner and risk owner; supporting sources: commissioning tests, inspections, incident records, work orders and affected activity owners. **Precedence:** the predefined success criteria and competent verification outrank favourable dashboard trends; formal new-risk findings outrank a narrow target metric. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** after the agreed observation period and on any adverse event. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: comparison and estimate. **Decision interval:** at the planned review point and monthly until stable. **Minimum evidence:** a defined intervention date, baseline, target metric, comparable observation window and quality-qualified data. **Method:** controlled before/after or interrupted time-series evaluation; estimate change with uncertainty, check seasonality and operational differences, and search for owner-defined unintended indicators. **Uncertainty:** A short baseline, concurrent changes, seasonality, proxy indicators and unobserved risk transfer limit attribution. **Safe failure:** Report Effect not attributable and continue monitoring or commission a stronger evaluation; do not close the risk from the indicator alone.

ANSWER BOUNDARY

A before-and-after association must not be presented as causal proof; competing changes, seasonality and altered reporting must remain visible. Do not claim causality from a short or uncontrolled period, certify effectiveness, ignore adverse trade-offs, infer personal outcomes or close the risk.

WHY THIS IS USEFUL

Decide whether to continue, adapt or investigate the control while commissioning stronger follow-up evidence where attribution is weak.

HEALTH AND SAFETY OFFICERS QUESTION

HS-036

Which evidence-supported conditions may have contributed to the event, and which suggested causes remain untested or contradicted?

SENSORS AND TELEMETRY

Proposed evidence may include permissioned incident timestamps, mapped alarm and asset-fault events, environmental measurements, work-order states and declared activity milestones. Use only the event window and relevant mapped zones or assets; retain raw timestamps and show any aggregate used. Evidence envelope: state modality and installation point, mapping version, sample interval, calibration or functional-test status, clock offset, uptime, gaps, drift or resets, permission and provenance. Do not inspect speech, research content, code or network content, and do not use sensor coincidence to infer causation.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the incident investigation lead owns the evidence plan; controlled sources include the incident record, witness accounts handled under authorised process, permits, maintenance and inspection records, asset logs and approved telemetry exports. **Precedence:** the investigation lead and competent evidence owners decide relevance; a telemetry correlation cannot override direct verified evidence. **Metadata:** show ID/version, owner, permission, effective or event time, retrieval time and chain of custody. **Staleness:** follow the investigation evidence plan. **Conflicts:** retain competing states; the investigation lead resolves them with the competent source owner.

ANALYSIS REQUIRED

Operation: observation, comparison and hypothesis screening. **Decision interval:** at investigation scoping and each evidence review. **Minimum evidence:** a bounded event definition, traceable timeline, at least one independent source for each proposed condition and an explicit list of missing evidence. **Method:** map observed conditions to credible mechanisms, test temporal order and contradictions, and classify each as supported, possible, unsupported or contradicted; do not calculate a causal probability. **Uncertainty:** Clock error, missing intervals, measurement limits, reporting bias and credible alternative explanations constrain every proposed contribution. **Safe failure:** Return Cause not established and request competent investigation rather than naming or ranking a cause.

ANSWER BOUNDARY

Do not assign blame, identify a person as the cause, infer intent or health, convert correlation into causation, or diagnose injury. Do not disclose protected witness or personal information, certify a conclusion, authorise work or control an asset.

WHY THIS IS USEFUL

Focus the competent investigation on evidence-supported conditions, reject contradicted explanations and record exactly what further inspection, testing or testimony is needed.

Incidents, near misses and corrective actions

HEALTH AND SAFETY OFFICERS QUESTION

HS-037

Which permitted logs, telemetry, photographs or controlled records may be lost under routine retention and need an authorised preservation decision now?

SENSORS AND TELEMETRY

Proposed retention-health metadata may show event-log age, overwrite cycles and available exports; it should not expand collection. **Timing:** use source-level retention clocks and the incident window. **Evidence quality:** show permissions, encryption, checksum/provenance, time synchronisation and unavailable sources; preserve only the minimum relevant scope. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the incident investigator, information-governance owner and each system owner; supporting sources: the approved retention schedule, legal/records advice and evidence-handling procedure. **Precedence:** an authorised preservation instruction and governance decision outrank automated recommendations; source owners control lawful export and chain of custody. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediately when evidence loss is plausible. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and recommendation. **Decision interval:** on incident escalation and before known overwrite points. **Minimum evidence:** an authorised case, defined time/location/asset scope and source retention metadata. **Method:** scoped preservation-risk assessment and evidence inventory; identify relevant sources, deletion deadlines, access constraints and minimal extract; produce a request, not an automatic hold. **Uncertainty:** Retention schedules, legal privilege, permissions, source ownership, event timestamps and chain-of-custody requirements may conflict. **Safe failure:** Alert the investigation and data owners before deletion as Preservation decision required; do not make an unauthorised copy.

ANSWER BOUNDARY

Do not impose a legal hold, copy restricted data without authority, widen surveillance, retain unrelated personal data or alter original evidence.

WHY THIS IS USEFUL

Issue a timely preservation request, protect chain of custody and avoid losing decision-critical evidence to routine deletion.

HEALTH AND SAFETY OFFICERS QUESTION

HS-038

For initial triage, which recorded severity and notification indicators are present, absent or still require competent or legal assessment?

SENSORS AND TELEMETRY

Proposed verified alarms, safety-service loss, active leak, route blockage, equipment fault and anonymous occupancy can supply factual indicators. **Timing:** use live events and 5-minute summaries. **Evidence quality:** show source reliability, location match, clock alignment, uptime and absence of coverage; no physiological, medical or emotion sensing is permitted. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the duty incident lead and emergency/technical responders; supporting sources: the original report, Security, first-aid and asset-owner records. **Precedence:** competent responder assessment and official emergency status outrank automated bands; reported facts remain distinct from telemetry. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediate receipt and refresh within 5 minutes during response. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and rule-based triage. **Decision interval:** immediately after report receipt and whenever verified facts change. **Minimum evidence:** current incident record, observed time, location/activity, recorded outcome indicators and the current notification criteria owned by the relevant authority. **Method:** test only explicit criteria, separate present, absent, unknown and requires competent assessment, and preserve the original report. **Uncertainty:** Facts may be incomplete or disputed and notification criteria can require legal interpretation that a rule engine cannot supply. **Safe failure:** Escalate for human triage whenever a time-critical criterion cannot be ruled out; do not issue a legal determination.

ANSWER BOUNDARY

Do not diagnose injury, assign statutory reportability, infer severity from absence of data, contact external bodies automatically or disclose personal details beyond authorised need. Do not certify compliance, assign blame or replace competent and legal judgement.

WHY THIS IS USEFUL

Prompt timely human escalation, preserve notification deadlines and show the officer which facts or legal judgement are still missing.

Incidents, near misses and corrective actions

HEALTH AND SAFETY OFFICERS QUESTION

HS-039

Have temporary controls and agreed lessons reached every affected team, shift, contractor and location within the required time?

SENSORS AND TELEMETRY

Proposed digital-signage heartbeat, message-delivery counts and briefing-record timestamps can show distribution, not reading or understanding. **Timing:** aggregate by team/location/shift and suppress small groups. **Evidence quality:** show audience mapping, failed delivery, clock alignment, system latency and offline channels; no individual engagement scoring. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the incident/action owner and safety-communications owner; supporting sources: distribution groups, shift plans, local briefing logs, training records and room ownership. **Precedence:** the approved audience and communication content outrank inferred recipients; local managers confirm practical briefing where required. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** within the temporary-control deadline and each shift change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation and comparison. **Decision interval:** on issue, at each shift handover and daily while temporary controls apply. **Minimum evidence:** approved message, affected scope, delivery channels and current team/location lists. **Method:** audience-impact mapping and communication-coverage reconciliation; compare required and evidenced coverage, identify undelivered or unbriefed groups and separate delivery from acknowledgement. **Uncertainty:** Recipient lists change, shifts and contractor teams rotate, acknowledgements can be delayed and receipt does not prove understanding. **Safe failure:** Return Reach not established, resend through approved channels and require supervisor confirmation for affected groups.

ANSWER BOUNDARY

Do not infer understanding, compliance or blame from delivery receipts, disclose personal engagement, or treat a digital acknowledgement as competence.

WHY THIS IS USEFUL

Re-brief missed teams and contractors, maintain temporary controls across shifts and record who owns remaining communication gaps.

HEALTH AND SAFETY OFFICERS QUESTION

HS-040

Is the evidence sufficient for the accountable owner to consider closing the corrective action, or are inspection, consultation, monitoring or verification still outstanding?

SENSORS AND TELEMETRY

Proposed follow-up telemetry may show sustained asset health or environmental improvement over a defined period. **Timing:** use the agreed review window with unchanged sampling where possible. **Evidence quality:** show calibration, coverage, clock alignment, downtime, baseline comparability and contrary signals; telemetry cannot replace consultation or physical verification. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the designated incident investigator and action/risk owners; supporting sources: competent inspection, consultation records, test results, CMMS and closure criteria. **Precedence:** formal owner sign-off against approved closure criteria outranks an automated completeness score; unresolved affected-person or technical concerns remain visible. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** at each closure review and after the agreed monitoring period. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** at action completion and formal case review. **Minimum evidence:** approved closure criteria, action evidence, owner verification and affected-party consultation where required. **Method:** closure-evidence completeness and residual-uncertainty assessment; trace each finding to action and evidence, identify open uncertainty and recommend inspection, consultation or monitoring. **Uncertainty:** Verification criteria, consultation, residual-risk review and post-action monitoring may be incomplete or conflicting. **Safe failure:** Keep the action open and state each outstanding inspection, consultation, measurement or owner decision.

ANSWER BOUNDARY

The service must not close an incident or action, accept residual risk or certify effectiveness. Do not close the case, certify that harm cannot recur, suppress dissenting evidence, infer acceptance from silence or lower residual risk automatically.

WHY THIS IS USEFUL

Give the accountable owner a bounded closure pack or keep the case open with a precise, proportionate evidence plan.

Contractors, permits and high-risk work

HEALTH AND SAFETY OFFICERS QUESTION

HS-041

Which contractor teams are expected on site in the decision window, and do their approved locations, supervision, induction and permit times align?

SENSORS AND TELEMETRY

Proposed access-control reporting may provide only aggregated contractor-credential counts by approved zone and time, not named movement histories. **Timing:** use event summaries at 15-minute intervals. **Evidence quality:** show reader coverage, anti-passback limits, clock alignment, tailgating uncertainty, uptime and permission scope; absence of a badge event is not absence from site. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the contractor manager, permit issuer and Security duty lead; supporting sources: work bookings, company/team register, inductions, escorts and access-control administration. **Precedence:** the approved permit and contractor booking outrank access telemetry; Security or permit-holder confirmation resolves operational presence. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 15 minutes for active work and the current permit version. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** before entry, at shift start and every 15 minutes for critical work. **Minimum evidence:** a valid company/team booking, authorised zone and permit time window. **Method:** permit-schedule-location reconciliation; compare planned and observed team-level activity cues, escorts and closures; flag unmatched zones without identifying workers. **Uncertainty:** Contractor rosters, permit times, work locations and supervisor cover can change after data retrieval; access or induction records do not prove readiness. **Safe failure:** Return Contractor readiness not confirmed and hold mobilisation for the contract or permit owner to resolve.

ANSWER BOUNDARY

Do not track named contractors, infer attendance or misconduct from badge data, disclose personal credential history, admit anyone or authorise work.

WHY THIS IS USEFUL

Resolve permit, location, supervision and access mismatches before entry, arrange escorts and prevent work in the wrong zone or time.

HEALTH AND SAFETY OFFICERS QUESTION

HS-042

Which planned high-risk work conflicts with public access, teaching, research, deliveries, emergency routes or another permit?

SENSORS AND TELEMETRY

Proposed anonymous occupancy, route flow, door state, loading-area and safety-service telemetry can confirm active interfaces. **Timing:** use 5-minute events and 15-minute summaries. **Evidence quality:** show spatial coverage, activity ambiguity, clock alignment, uptime and blind spots; no content inspection or person identification. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the permit issuer, area owner and event/timetabling owners; supporting sources: other permits, delivery bookings, closure plans, Security and Estates schedules. **Precedence:** a formal permit, isolation or closure instruction outranks calendars and telemetry; the competent permit issuer resolves conflicts. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before issue and every 15 minutes during high-risk work. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: comparison and recommendation. **Decision interval:** at application, issue, shift handover and active work. **Minimum evidence:** two current activities or permits sharing a zone, route, service or risk interface. **Method:** spatial-temporal permit conflict and interface analysis; apply approved separation, exclusion and service-dependency rules; expose unknown activity and unmapped zones. **Uncertainty:** Schedules can slip, permit boundaries may be imprecise and route or closure states may change during the planned work window. **Safe failure:** Return Work-interface conflict not ruled out and require permit owners to separate work, routes or time windows.

ANSWER BOUNDARY

Do not inspect research content, cancel events, issue or revoke a permit, declare interfaces safe, or directly control access and services.

WHY THIS IS USEFUL

Reschedule, segregate, isolate or coordinate incompatible work before public, teaching, research, delivery or emergency-route conflicts occur.

Contractors, permits and high-risk work

HEALTH AND SAFETY OFFICERS QUESTION

HS-043

Do the current isolation record, permit boundary and independently verified asset state agree for the exact electrical, mechanical, gas or other system?

SENSORS AND TELEMETRY

Proposed breaker, valve, lock-sensor or control-system position may corroborate an approved isolation only where designed and commissioned. **Timing:** use change events and 1-minute heartbeat. **Evidence quality:** show asset identity, sensing point, fail-state logic, clock alignment, maintenance mode and blind spots; remote indication never proves zero energy. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the authorised isolating person and permit issuer; supporting sources: isolation certificate, asset drawings, lock register, CMMS and control-system records. **Precedence:** the authorised person's physical isolation, test and documented certificate outrank any remote indication; the current permit scope governs. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** live during issue, transfer and cancellation. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and comparison. **Decision interval:** at every permit state change and before handover. **Minimum evidence:** a current permit, isolation certificate and asset identifiers; telemetry is optional corroboration. **Method:** permit-to-isolation scope and state reconciliation; match energy sources, isolation points, locks, tests and work boundary; treat any mismatch or unmapped point as unsafe to rely on. **Uncertainty:** Asset labels may be wrong, remote state cannot prove zero energy and multiple stored or secondary energy sources may be omitted. **Safe failure:** Return Isolation not independently verified and prohibit reliance on the answer until competent lockout, test and permit confirmation.

ANSWER BOUNDARY

Telemetry must not create, prove or cancel an isolation, and the service must not authorise energisation or work. Do not confirm dead or zero-energy state, operate breakers or valves, release locks, issue a permit, or substitute telemetry for test-before-touch.

WHY THIS IS USEFUL

Pause the work-control decision, reconcile the isolation boundary with the technical owner and obtain independent verification before authorisation.

HEALTH AND SAFETY OFFICERS QUESTION

HS-044

Does the hot-work plan cover combustible control, alarm impairment, fire watch, handback and post-work monitoring for the full risk window?

SENSORS AND TELEMETRY

Proposed area temperature, smoke/heat detector availability and fire-system impairment status can provide context; no sensor replaces a fire watch. **Timing:** use 1-minute events during work and the approved post-work period. **Evidence quality:** show placement, alarm masking, calibration, clock alignment, coverage and uptime. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the hot-work permit issuer and fire-safety responsible person; supporting sources: method statement, fire-watch rota, alarm impairment log, area closure and contractor supervisor. **Precedence:** the approved hot-work permit and competent fire-safety instruction outrank telemetry; formal impairment and watch records define the control window. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before ignition, continuously during work and through the specified post-work period. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at permit issue, start, pauses, completion and monitoring end. **Minimum evidence:** a valid permit, work location/time, required controls, named responsible roles and current alarm status. **Method:** time-window control-completeness and fire-system dependency checking; verify coverage across the whole risk window, identify gaps at handover and require human confirmation of housekeeping and watch. **Uncertainty:** Local combustibles can be unseen, alarm impairment status may change and fire-watch or post-work coverage may not span the full risk period. **Safe failure:** Return Hot-work controls incomplete or unknown and refer the hold decision to the permit issuer; never authorise the work.

ANSWER BOUNDARY

Do not issue the permit, act as fire watch, detect all ignition sources, declare the area fire-safe, silence alarms or authorise work during an impairment.

WHY THIS IS USEFUL

Close fire-watch, alarm-impairment and post-work monitoring gaps, then keep the permit open for the full risk window.

Contractors, permits and high-risk work

HEALTH AND SAFETY OFFICERS QUESTION

HS-045

Are the access equipment, edge protection, anchor arrangements, weather limits and exclusion zone documented and available for the planned work at height?

SENSORS AND TELEMETRY

No fixed building sensor can confirm safe work-at-height equipment or anchors. Proposed evidence may include current equipment inspection tags, asset-register status, approved roof or access-point weather readings and temporary exclusion-zone contacts where explicitly commissioned. Cover only the named equipment, anchor or work zone and permit window; weather summaries must retain source and update time. Evidence envelope: show modality, exact asset/location, mapping, interval, inspection or calibration state, clock offset, uptime, gaps, drift, permission and provenance. A pre-use physical check remains mandatory.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the work-at-height permit/task owner and competent equipment inspector; supporting sources: method statement, equipment register, inspection tags, exclusion plan and weather source. **Precedence:** a competent pre-use/inspection decision and approved work plan outrank scans or barrier telemetry; the current permit defines acceptable conditions. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before each shift and after alteration or adverse conditions. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at permit issue, shift start and condition changes. **Minimum evidence:** current work plan, equipment/inspection evidence and location-specific controls. **Method:** work-at-height readiness and exclusion-zone completeness check; match access equipment, edge protection, rescue, exclusion and environmental conditions; return inspection required for any unverified element. **Uncertainty:** Anchor condition, equipment compatibility, local weather, forecast error and exclusion-zone integrity require site-specific competent judgement. **Safe failure:** Return Work-at-height readiness unknown and hold mobilisation for competent equipment, anchor and weather verification.

ANSWER BOUNDARY

Do not certify a ladder, scaffold or anchor, infer safe weather from one sensor, authorise access, inspect workers by camera or control barriers.

WHY THIS IS USEFUL

Hold mobilisation, replace unsuitable equipment, revise the exclusion zone or weather window, and obtain competent verification of anchors and access arrangements.

HEALTH AND SAFETY OFFICERS QUESTION

HS-046

Is entry to the named plant room, roof, riser or other restricted area supported by current access, supervision, communication and emergency arrangements?

SENSORS AND TELEMETRY

Proposed door state, approved environmental/gas alarms and anonymous zone presence may support readiness only where those systems are designed for the space. **Timing:** use events and 1-minute health. **Evidence quality:** show alarm type, placement, calibration, clock alignment, access-reader coverage and uptime; no sensor classifies the space or guarantees atmosphere. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the restricted-space owner, permit issuer and Security; supporting sources: access authorisation, method statement, rescue/emergency plan, asset status and supervision arrangements. **Precedence:** the competent owner's classification and permit/access decision outrank telemetry; formal closure or alarm fault prevents reliance. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before entry and continuously for approved alarms during occupation. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at request, entry, handover and exit. **Minimum evidence:** current space rules, authorised task, required supervision/rescue and valid access/permit records. **Method:** entry-gate readiness and emergency-arrangement matching; check each prerequisite and active impairment; use telemetry only to flag contradiction or loss of a required system. **Uncertainty:** An access record does not prove permission or presence, communications may have dead spots and rescue arrangements may not cover the exact zone. **Safe failure:** Return Restricted-entry support unresolved and refer the entry decision to the area and permit owners.

ANSWER BOUNDARY

Do not classify a confined or restricted space legally, authorise entry, guarantee atmosphere, identify entrants from sensor data or unlock the door.

WHY THIS IS USEFUL

Block unsupported entry, arrange supervision, communication and rescue, and confirm local controls before access is granted.

Contractors, permits and high-risk work

HEALTH AND SAFETY OFFICERS QUESTION

HS-047

Does the lifting operation, large delivery or materials move remain within load, route, exclusion, weather and emergency-access constraints?

SENSORS AND TELEMETRY

Proposed lift load/availability, loading-bay status, door clearances and anonymous route counts can corroborate logistics. **Timing:** use event state and 5-minute route summaries. **Evidence quality:** show sensor range, scale calibration, route coverage, clock alignment, uptime and object-size limits; telemetry cannot confirm load weight, stability or rigging. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the lifting/delivery plan owner and relevant Estates/structural asset owner; supporting sources: route plan, lift plan, delivery booking, equipment certificates, closure plan and Security. **Precedence:** the approved lift/delivery plan and competent capacity confirmation outrank sensor estimates; formal route closure outranks observed low occupancy. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before movement and every 5 minutes during the operation. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation, forecast and comparison. **Decision interval:** at planning, pre-start and during critical movement. **Minimum evidence:** approved load/size, route, equipment and current closure/access states. **Method:** route, capacity and interface constraint checking; compare dimensions, capacities, turns, lift availability, pedestrian conflict and emergency-route dependencies; require direct confirmation for unknown loads. **Uncertainty:** Actual load, centre of gravity, route dimensions, floor capacity, weather and emergency-route availability may differ from the plan. **Safe failure:** Return Move not demonstrated within constraints and require a competent lifting or logistics plan before proceeding.

ANSWER BOUNDARY

A load or route calculation and weather forecast must not replace a competent lifting plan, structural check or site decision. Do not certify load weight, rigging, structural capacity or lift suitability, direct the operation, identify passers-by or control lifts and doors.

WHY THIS IS USEFUL

Change route, load, timing or exclusion, protect emergency access and obtain technical approval before the move.

HEALTH AND SAFETY OFFICERS QUESTION

HS-048

Are hazardous substances, cylinders, batteries or waste covered by current approval, inventory, storage, segregation, ventilation, spill and disposal arrangements?

SENSORS AND TELEMETRY

Proposed cabinet/door state, storage temperature, gas or leak alarms, cylinder-restraint contacts and inventory weight may cover selected stores. **Timing:** use 1-minute alarms and 15-minute summaries. **Evidence quality:** show substance-specific sensor limits, calibration, clock alignment, coverage, uptime and uninstrumented containers; sensors do not identify unknown contents. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the substance/task owner and hazardous-store or waste owner; supporting sources: approved inventory, safety data sheets, substance risk assessment, delivery record, permit and waste documentation. **Precedence:** the approved assessment and labelled inventory outrank sensor inference; competent store-owner instructions and formal segregation rules govern. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before delivery/use and within 1 minute for verified alarms. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at booking, receipt, storage change and task start. **Minimum evidence:** identified material, quantity, approved location, required controls and current store status. **Method:** inventory-location-time and control-readiness screening; check compatibility, capacity, ventilation/alarm availability, expiry and waste route; mark unknown material or missing documents as not ready. **Uncertainty:** Inventory accuracy, substance identity and quantity, container integrity, segregation, ventilation and waste classification may all be uncertain. **Safe failure:** Return Substance control not verified and quarantine or hold through the responsible owner pending competent advice.

ANSWER BOUNDARY

Do not identify an unknown chemical from sensor data, declare storage safe, interpret exposure medically, accept a delivery or authorise use/disposal.

WHY THIS IS USEFUL

Prepare approved storage, segregation, ventilation, spill and disposal controls; reject or quarantine unsupported materials before entry.

Contractors, permits and high-risk work

HEALTH AND SAFETY OFFICERS QUESTION

HS-049

Before mobilisation, are contractor induction, task competence, supervision, method statement and permit dependencies current and traceable?

SENSORS AND TELEMETRY

No sensor can prove competence or understanding. Proposed system status may only show that a controlled record was issued, acknowledged or expired. **Timing:** use task-level status, not individual monitoring. **Evidence quality:** show record version, verification method, permission, update time and missing external certificates; access events are not competence evidence. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the contractor manager, permit issuer and task supervisor; supporting sources: learning records, competence certificates, approved method statements, company assurance and induction logs. **Precedence:** the permit issuer's verified task-specific decision and current controlled documents outrank completion flags; practical supervision needs cannot be inferred from certificates. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before mobilisation, permit issue and role change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at prequalification, mobilisation and each permit issue. **Minimum evidence:** defined task/role requirements plus current verified records and supervision plan. **Method:** task-role-document completeness and expiry checking; match required induction, competence, method, emergency and local familiarisation; distinguish document present, verified, expired and practically unassessed. **Uncertainty:** Recorded induction does not prove task competence; team, method, permit scope, authorisation and supervision may change before mobilisation. **Safe failure:** Return Contractor readiness not established and do not mobilise until the responsible owner verifies every dependency.

ANSWER BOUNDARY

Do not declare a person competent solely from records, disclose unnecessary personal data, infer ability from past access, issue a permit or replace supervisor judgement.

WHY THIS IS USEFUL

Hold mobilisation, obtain missing competence or method evidence, add supervision and assign only authorised roles.

HEALTH AND SAFETY OFFICERS QUESTION

HS-050

Does the handback evidence support the authorised owner in deciding whether the area or asset may reopen after contractor work?

SENSORS AND TELEMETRY

Proposed asset, alarm, guard/interlock, door and environmental states can support handback at mapped points. **Timing:** use event state plus an agreed stability period. **Evidence quality:** show commissioning/test mode, calibration, clock alignment, sensor coverage, uptime and contrary alarms; normal telemetry cannot prove tools, debris or temporary controls are removed. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the contractor supervisor, technical asset owner and area owner; supporting sources: handback certificate, test results, work order, permit cancellation, inspection and Security closure record. **Precedence:** competent owner inspection and signed handback outrank telemetry or contractor self-declaration; formal closure remains until the authorised owner removes it. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** at work completion and before reopening. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation, comparison and recommendation. **Decision interval:** at completion, permit cancellation and proposed reopening. **Minimum evidence:** defined acceptance criteria, signed contractor evidence, owner inspection and relevant asset/system states. **Method:** handback checklist, state reconciliation and stability-window review; trace each criterion, identify unverified physical conditions and flag any active fault, isolation or temporary control. **Uncertainty:** Hidden defects, temporary services, cleanliness, incomplete testing and unresolved ownership can remain after a handback record is signed. **Safe failure:** Return Handback incomplete, keep the area or asset restricted and list the verification needed by the reopening owner.

ANSWER BOUNDARY

The service must not reopen an area, remove a restriction, accept handback or authorise use. Do not commission or reopen the area, certify workmanship, cancel a permit, reset alarms, remove isolation or accept housekeeping without inspection.

WHY THIS IS USEFUL

Give the authorised owner a handback exception list covering unresolved defects, missing tests, housekeeping and restoration evidence.

Training, competence and communications

HEALTH AND SAFETY OFFICERS QUESTION

HS-051

Which safety-critical roles lack current required training for the assigned duty, equipment or operating period?

SENSORS AND TELEMETRY

No sensor can establish training or competence. Proposed record integration may report role-level complete, due, expired or unavailable status, with small-group suppression. **Timing:** use daily snapshots. **Evidence quality:** show role mapping, course version, external-record gaps, permission and synchronisation age; never combine with location or performance telemetry. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the learning-system owner, role owner and local manager; supporting sources: HR role assignments, duty rotas, external certificates and the safety training matrix. **Precedence:** the current approved role requirement and verified completion record outrank inferred competence; the manager confirms actual assignment. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** one day for routine status and before a critical duty starts. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** daily, at rota publication and before high-risk work. **Minimum evidence:** a current assigned role, approved requirement set and permissioned completion status. **Method:** role-training requirement matching and coverage-gap detection; compare validity dates, course versions, equivalence and role changes; distinguish not trained, expired, evidence unavailable and not required. **Uncertainty:** Training matrices can lag role changes, course equivalence may be unclear and expiry or duty scope may not match the assignment. **Safe failure:** Return Training status unknown and remove the role from unsupervised assignment until the training owner confirms it.

ANSWER BOUNDARY

Do not infer practical competence, expose individual records beyond need-to-know, rank staff performance, remove a role or authorise duty.

WHY THIS IS USEFUL

Schedule required training, adjust duty cover and prevent assignment where the current record is missing, expired or out of scope.

HEALTH AND SAFETY OFFICERS QUESTION

HS-052

Are current authorisations and competence records in place for the people assigned to the named equipment or high-risk task?

SENSORS AND TELEMETRY

No sensor or equipment-use log proves competence. Proposed integration may show whether a current authorisation record exists for the task, without publishing unrelated training history. **Timing:** use point-in-time checks at assignment. **Evidence quality:** show verification source, expiry, permission, identity-resolution quality and unavailable external evidence. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the equipment/task authorisation owner and line manager; supporting sources: learning records, practical assessments, supervisor sign-off and contractor competence evidence. **Precedence:** the current task-specific authorisation and competent assessor decision outrank generic training completion or previous equipment use. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** at assignment, permit issue and authorisation change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** before each high-risk assignment and daily for active rosters. **Minimum evidence:** an authorised identity lookup, task/equipment ID, required competence and current authorisation evidence. **Method:** person-role-task authorisation validation; check scope, expiry, restrictions and supervisor conditions; return permission denied or unverified rather than inferring from history. **Uncertainty:** A record may not cover the exact asset or task, practical competence may have lapsed and required supervision may be missing. **Safe failure:** Return Authorisation or competence not confirmed and require the owner to change the assignment or arrange assessment and supervision.

ANSWER BOUNDARY

Do not infer competence from usage, access or incident history, disclose full personnel records, authorise the task, or create a personal risk score.

WHY THIS IS USEFUL

Reassign, supervise, assess or renew authorisation before the named person undertakes the equipment or high-risk task.

Training, competence and communications

HEALTH AND SAFETY OFFICERS QUESTION

HS-053

Which roles need a refresher or change-specific briefing because of a revised control, incident lesson, equipment change or long period without practice?

SENSORS AND TELEMETRY

No sensor determines learning need. Proposed change feeds may identify affected assets, locations, roles and shifts; communication systems may show delivery status only. **Timing:** use change-event records and daily audience snapshots. **Evidence quality:** show impact mapping, version, clock alignment, permission and missing contractors or visitors. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the policy/change owner and training or communications owner; supporting sources: role matrices, room/asset ownership, incident lessons, rotas and contractor scopes. **Precedence:** the approved change impact and required audience outrank inferred users; local managers confirm who performs the affected task. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** at change approval and before the changed arrangement goes live. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation and recommendation. **Decision interval:** at each approved change and daily until coverage is complete. **Minimum evidence:** a controlled change, effective date, affected tasks/locations and current audience records. **Method:** change-to-audience impact mapping and training-gap analysis; map direct and interface roles, identify old-version completion and unbriefed shifts, and separate formal training from local briefing. **Uncertainty:** Change notices and incident lessons may not map cleanly to roles, and inactivity thresholds or refresher rules may differ by task. **Safe failure:** Return Refresher need unresolved and route the role-task pair to the training and activity owners before duty.

ANSWER BOUNDARY

Do not infer who understood the change, inspect work content, disclose personal learning history, or declare competence after message delivery.

WHY THIS IS USEFUL

Target only the affected roles, deliver change-specific briefing and stop superseded controls continuing through generic training records.

HEALTH AND SAFETY OFFICERS QUESTION

HS-054

Does every operating period have the owner-approved minimum cover for first aid, fire response, incident coordination and other emergency roles?

SENSORS AND TELEMETRY

No sensor can establish competence, willingness or actual availability. Proposed rota and check-in integrations may show role-level planned and confirmed cover; access events may only flag a discrepancy for human confirmation and must not be used as presence proof. **Timing:** assess each operating period and update on approved rota or check-in change. **Evidence quality:** show rota age, check-in state, role scope, location coverage, permission, integration health and unresolved absences without movement history. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the emergency-role owner and local duty manager; supporting sources: verified competence records, approved rotas, bookings, planned events and authorised absence updates. **Precedence:** the current duty rota and manager confirmation outrank historical assignments or badge presence; role competence records define eligibility. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 15 minutes for active coverage and at every rota change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation and comparison. **Decision interval:** at rota publication, daily opening and before events. **Minimum evidence:** approved minimum coverage, current qualified-role roster and planned activity/occupancy context. **Method:** time-zone emergency-role coverage and contingency analysis; calculate covered, marginal and uncovered periods, test single absence and identify approved escalation or shared coverage. **Uncertainty:** A rota does not prove availability, one person may hold conflicting roles and concurrent events can consume planned cover. **Safe failure:** Return Required cover not confirmed and ask the duty manager to adjust opening, staffing or the activity plan.

ANSWER BOUNDARY

Do not guarantee that a named person is physically present, reveal absence or health information, infer willingness, assign the role or certify coverage adequacy.

WHY THIS IS USEFUL

Adjust rotas, share approved cover, change operating arrangements and brief the duty lead before a minimum-role gap occurs.

Training, competence and communications

HEALTH AND SAFETY OFFICERS QUESTION

HS-055

Which critical safety communications have not reached all affected teams, contractors or locations, or lack the required acknowledgement?

SENSORS AND TELEMETRY

Proposed message-delivery receipts, display heartbeat and local briefing timestamps can show channel distribution, not reading or understanding. **Timing:** aggregate by group, shift and location. **Evidence quality:** show audience-list age, failed addresses, offline channels, clock alignment, permission and small-group suppression; no individual engagement score. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the safety-communications owner and instruction owner; supporting sources: distribution groups, shift plans, room ownership, contractor contacts and local briefing logs. **Precedence:** the approved instruction and audience list outrank inferred recipients; local managers validate non-digital briefing needs. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** at issue, each shift change and before the instruction takes effect. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation and comparison. **Decision interval:** near-live after issue and daily while active. **Minimum evidence:** controlled message version, affected audience/location and authorised delivery evidence. **Method:** communication coverage and delivery-exception reconciliation; compare required channels and groups, identify failed or unrecorded delivery, and separate sent, delivered, acknowledged and briefed. **Uncertainty:** Distribution lists can be stale, contractors or shifts omitted, acknowledgements delayed and receipt does not demonstrate comprehension. **Safe failure:** Return Communication reach unknown, resend in approved channels and obtain supervisor confirmation where the consequence is high.

ANSWER BOUNDARY

Do not infer comprehension, compliance or blame, disclose personal engagement data, alter the instruction or treat acknowledgement as training.

WHY THIS IS USEFUL

Reissue through appropriate channels, brief missed shifts and contractors, and preserve a traceable communication record.

HEALTH AND SAFETY OFFICERS QUESTION

HS-056

Which signs, labels, quick-reference instructions and emergency notices are missing, outdated, illegible, inaccessible or inconsistent with current arrangements?

SENSORS AND TELEMETRY

Proposed digital-display heartbeat and local illuminance may support electronic or visibility checks; physical signs still need inspection. **Timing:** use daily status and inspection timestamps. **Evidence quality:** show sign/register mapping, display version, viewing-position limits, clock alignment, uptime and inaccessible areas; no camera-based reading of notices or people. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the sign/instruction owner and local area owner; supporting sources: controlled sign register, accessibility guidance, fire/emergency plans, inspection records and change control. **Precedence:** the current approved sign content and competent placement/accessibility inspection outrank display telemetry or old floor-plan locations. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** daily for digital status and after any route, hazard or policy change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation and comparison. **Decision interval:** daily for critical signs and at each change. **Minimum evidence:** a current sign requirement, location, approved version and recent inspection or display status. **Method:** sign-version, placement and inspection-completeness checking; match required signs to actual locations and versions, identify obscured/uninspected positions and route accessibility questions to consultation. **Uncertainty:** The approved sign schedule may not match the observed location, and lighting, obstruction, language or temporary changes can affect usability. **Safe failure:** Return Signage status not verified and arrange a physical check plus an approved temporary notice if needed.

ANSWER BOUNDARY

Do not certify accessibility or legibility for every user, infer disability, use camera surveillance, or replace a competent physical inspection.

WHY THIS IS USEFUL

Replace or reposition signs, correct conflicting instructions and make emergency and hazard information accessible before the area is used.

Training, competence and communications

HEALTH AND SAFETY OFFICERS QUESTION

HS-057

Does the recorded pre-start briefing cover the actual task hazards, recent changes, interfaces, stop conditions and emergency actions?

SENSORS AND TELEMETRY

No sensor can prove the quality of a briefing. Proposed records may show the controlled briefing version, task, time, team and acknowledgement status. **Timing:** use task-level snapshots. **Evidence quality:** show document version, attendee-verification method, permission and missing off-system participants; do not record speech or analyse attention. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the task supervisor and permit or work owner; supporting sources: method statement, risk assessment, change notices, interface plans and briefing record. **Precedence:** the supervisor's task-specific confirmation and current controlled documents outrank a generic attendance tick or old briefing. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediately before work and after material change or shift handover. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at pre-start and every material change. **Minimum evidence:** current task controls, stop conditions, affected interfaces and authorised briefing evidence. **Method:** briefing-content and audience-completeness checking; verify that required topics and roles are covered, identify superseded content and require human confirmation of questions and understanding. **Uncertainty:** The task may change after briefing, attendance may be incomplete and a signed record does not show understanding of stop conditions. **Safe failure:** Return Pre-start briefing not demonstrated and require the supervisor to repeat and record the task-specific briefing.

ANSWER BOUNDARY

Do not record or analyse speech, infer attention or understanding, declare competence, issue a permit or authorise the start of work.

WHY THIS IS USEFUL

Pause and re-brief the team, clarify interfaces and stop conditions, and close task-specific omissions before work begins.

HEALTH AND SAFETY OFFICERS QUESTION

HS-058

Are required formats, languages and accessibility provisions available for the safety communication without collecting or inferring sensitive personal conditions?

SENSORS AND TELEMETRY

No sensor may infer language, disability, literacy or health. Proposed channel analytics may report aggregate delivery success by approved audience group and format. **Timing:** use campaign-level summaries with suppression. **Evidence quality:** show consent/purpose, audience-source provenance, permissions, update age and missing offline channels; no behavioural profiling. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the communication owner and authorised accessibility/inclusion service; supporting sources: person-provided preferences, workforce/student communication plans, contractor arrangements and local managers. **Precedence:** explicit person-provided or approved group requirements outrank inferred preferences; medical or adjustment details remain in restricted systems. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the person-agreed review date or before each critical communication. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** during communication design and after delivery exceptions. **Minimum evidence:** approved content, authorised audience requirements and available formats/channels. **Method:** privacy-preserving requirement-to-format and channel-coverage mapping; check plain-language, alternative-format, language and non-digital routes without exposing why a format is needed. **Uncertainty:** Communication needs must not be inferred, and translation quality, format availability or assistive access may not be independently verified. **Safe failure:** Return Accessible communication not confirmed and offer approved formats while consulting the accessibility owner.

ANSWER BOUNDARY

Do not infer disability, language or capability from behaviour or device settings, reveal adjustment reasons, translate safety-critical meaning without competent review, or claim understanding.

WHY THIS IS USEFUL

Provide accessible formats, interpretation or human support while keeping sensitive personal information outside communication analytics.

Training, competence and communications

HEALTH AND SAFETY OFFICERS QUESTION

HS-059 Which exercise, incident or inspection findings have not yet been incorporated into training content, scenarios, role cards or local instructions?

SENSORS AND TELEMETRY

Proposed drill alarm, door and anonymous zone-count timestamps can corroborate sequence and system behaviour. **Timing:** use event-window data and aggregated timing. **Evidence quality:** show clock alignment, coverage, sensor mode, uptime and gaps; no individual performance tracking. Training-system data should show version and audience, not personal scoring. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the emergency-planning, training and inspection-programme owners; controlled sources include exercise reports, incident lessons, inspection findings, action registers, approved course content, role cards and local instructions. **Precedence:** formally accepted findings and current approved materials outrank draft lessons, attendance records or telemetry. **Metadata:** show record ID/version, decision owner, effective and event times, retrieval time, permissions and lineage. **Staleness:** apply each action due date and document review date. **Conflicts:** retain competing states; the finding owner and training owner resolve them jointly.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** weekly after a debrief and at closure review. **Minimum evidence:** approved findings, actions, affected roles and current training/instruction versions. **Method:** finding-to-action-to-training traceability analysis; trace each lesson to an owner, change and audience; flag unimplemented, untested and superseded items. **Uncertainty:** A finding may not be formally accepted, the training version may lag and completion data may not show whether the revised learning works. **Safe failure:** Keep the lesson action open and route the missing content, rollout or effectiveness test to the finding and training owners.

ANSWER BOUNDARY

Do not score individual drill performance, infer panic or competence, alter emergency procedures without approval, or declare training effective from document updates.

WHY THIS IS USEFUL

Convert accepted exercise, incident and inspection lessons into revised learning objectives, scenarios, role cards and a test of whether the change works.

HEALTH AND SAFETY OFFICERS QUESTION

HS-060 Where would a training-completion record overstate readiness because practical competence, familiarisation, supervision or observed performance still needs human verification?

SENSORS AND TELEMETRY

No sensor or usage history proves practical competence. Proposed systems may compare task requirements with records of course, supervised practice, authorisation and local induction. **Timing:** use point-in-time status. **Evidence quality:** show source type, assessor, version, expiry, permission and unavailable evidence; exclude productivity, error-rate and personal-surveillance data. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the competence/authorisation owner and task supervisor; supporting sources: learning records, practical assessments, local induction, supervised practice and incident restrictions. **Precedence:** a competent assessor or supervisor decision outranks course completion, self-declaration or equipment usage; current restrictions remain authoritative. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** before first assignment, after change and at authorisation renewal. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at assignment and periodic authorisation review. **Minimum evidence:** task-specific competence criteria and current evidence across knowledge, practical assessment, local familiarity and supervision. **Method:** evidence-sufficiency classification for competence decisions; classify documented, practically verified, supervised-only, expired, unavailable and conflicting evidence; do not derive a score. **Uncertainty:** Certificates do not prove current practical skill, assessor validity, task familiarity or performance under the required supervision level. **Safe failure:** Return Readiness not established and require practical assessment, familiarisation or supervised work.

ANSWER BOUNDARY

Do not infer practical competence, understanding or safe performance from attendance, quiz results or completion records. Do not rank individuals, infer competence or health from performance, disclose unrelated records, authorise work or replace observation by a competent supervisor.

WHY THIS IS USEFUL

Target practical assessment, supervision or familiarisation where a completion record does not establish safe readiness.

Emergency preparedness and welfare arrangements

HEALTH AND SAFETY OFFICERS QUESTION

HS-061

Which fire detection, alarm, voice notification, emergency lighting and controlled door-release functions are available, impaired, in test or unknown for their intended zones?

SENSORS AND TELEMETRY

Proposed approved panel integrations may expose alarm-system health, zone faults, test mode, battery status, emergency-light test status and door-release feedback. **Timing:** use event states and 1-minute heartbeat. **Evidence quality:** show zone mapping, commissioning, diagnostic coverage, clock alignment and maintenance impairments; remote health cannot prove audibility or illumination. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the fire-safety responsible person and each safety-system asset owner; supporting sources: maintenance/test records, impairment logs, Security and current emergency plan. **Precedence:** the official system panel and competent test/inspection decision outrank a secondary dashboard; formal impairment instructions govern operation. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 1 minute for active health and the latest completed test/inspection. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and comparison. **Decision interval:** live to 1 minute and at daily opening. **Minimum evidence:** current authoritative system state, zone mapping and active impairment records. **Method:** safety-system health, zone coverage and dependency evaluation; identify faults, disabled zones, failed tests and dependencies, while separating not monitored from healthy. **Uncertainty:** A secondary dashboard may lag the panel, zone mapping may be wrong and remote health cannot prove audibility, illumination or field performance. **Safe failure:** Return Safety-system function unknown or impaired, escalate to the asset owner and apply only approved temporary arrangements.

ANSWER BOUNDARY

System status must not be presented as a fire-safety certificate or used to control alarm, release or evacuation functions. Do not certify fire-system compliance, silence or reset alarms, initiate tests, unlock doors, promise audibility or order building occupation/evacuation.

WHY THIS IS USEFUL

Escalate impairments, apply only approved temporary arrangements and restrict affected zones until the service owner confirms restoration.

HEALTH AND SAFETY OFFICERS QUESTION

HS-062

Do current evacuation routes, refuges, assembly points and temporary diversions agree across approved plans, signs, closures and accessibility arrangements?

SENSORS AND TELEMETRY

Proposed door, lift, refuge-call and selected route-obstruction states can corroborate current availability. **Timing:** use 5-minute status and change events. **Evidence quality:** show exact route coverage, sensor limits, clock alignment, uptime and unmonitored obstacles; anonymous counts may support demand only and must not identify assistance needs. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the emergency-plan and fire-safety owners; supporting sources: current route/assembly drawings, Security closures, Estates faults, event plans and accessibility arrangements. **Precedence:** the approved emergency plan and competent temporary instruction outrank signage scans, historical plans or telemetry; person-specific plans remain restricted. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediately after any closure/change and every 5 minutes for critical asset loss. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and comparison. **Decision interval:** at daily opening, event setup and every route change. **Minimum evidence:** current plan, route/assembly records, closure status and relevant asset state. **Method:** spatial consistency and route-dependency reconciliation; compare plans, signs, diversions and dependencies, identify contradictions and require physical confirmation of clearance. **Uncertainty:** Plans, signs, closures and diversions may have different versions, while physical route and accessibility conditions can change without telemetry. **Safe failure:** Return Route agreement not established and require competent inspection and owner correction before occupation.

ANSWER BOUNDARY

Do not disclose personal evacuation plans, certify a route, direct an emergency evacuation, infer disability or automatically operate doors and lifts.

WHY THIS IS USEFUL

Correct plans, signs, diversions and briefings, deploy stewards where needed and resolve accessibility conflicts before occupancy.

Emergency preparedness and welfare arrangements

HEALTH AND SAFETY OFFICERS QUESTION

HS-063

Is the approved first-aid cover and equipment provision suitable for the declared occupancy, activities, hazards and operating period?

SENSORS AND TELEMETRY

No sensor can decide whether first-aid provision is adequate. Proposed cabinet or kit status may report seal, stock-check, expiry or defibrillator self-test states, while privacy-approved occupancy counts may support demand context. **Timing:** use current equipment events and 15-minute zone totals for the declared activity window. **Evidence quality:** show kit mapping, inspection date, device-test limitations, count uncertainty, small-group suppression, clock alignment, uptime and unmonitored locations. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the first-aid needs-assessment owner and local duty manager; supporting sources: qualified-role rota, equipment inspection log, event/activity plan and supplier/service records. **Precedence:** the current needs assessment and competent duty decision outrank occupancy ratios or device telemetry; verified responder qualification defines eligibility. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** daily opening, before events and immediately on equipment fault. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation, comparison and recommendation. **Decision interval:** daily and before planned peaks. **Minimum evidence:** approved needs criteria, qualified rota, planned activity/occupancy and current equipment inspection status. **Method:** time-location first-aid resource coverage and readiness checking; identify uncovered periods, unavailable equipment and travel-distance concerns; test reasonable single-failure scenarios. **Uncertainty:** Needs-assessment assumptions, duty rosters, response time, equipment contents and activity risk may change independently. **Safe failure:** Return First-aid adequacy unresolved and ask the duty manager and first-aid owner to adjust cover, equipment or the activity.

ANSWER BOUNDARY

A staffing or stock calculation must not replace the first-aid needs assessment or disclose health information. Do not give medical advice, diagnose need, guarantee responder presence or response time, disclose health events, allocate a role or certify equipment.

WHY THIS IS USEFUL

Adjust first-aid cover, relocate or replenish equipment and change the activity plan through the responsible owner.

HEALTH AND SAFETY OFFICERS QUESTION

HS-064

Are emergency contacts, duty managers and escalation routes current, reachable and authorised for this location and time?

SENSORS AND TELEMETRY

No sensor determines who should be contacted. Proposed roster and communications-system health may show duty coverage and channel availability. **Timing:** use 15-minute duty windows and live service status. **Evidence quality:** show last update, role-zone mapping, permission, failed channels and backup route; do not publish personal details beyond authorised users. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the emergency-planning owner and duty-rota owner; supporting sources: Security, Estates, first-aid, senior management and service-provider contact registers. **Precedence:** the current approved duty rota and emergency call tree outrank directory entries, calendars or historical documents. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 15 minutes for duty coverage and immediately after a contact change. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and observation. **Decision interval:** at daily opening, shift handover and incident activation. **Minimum evidence:** current operational period, location, incident type and authorised contact/role records. **Method:** time-zone escalation-chain validation and fallback-path checking; trace primary and backup routes, identify unreachable or vacant roles and preserve need-to-know disclosure. **Uncertainty:** Contact records and rotas can lag, a device may be unreachable and delegation or out-of-hours fallback may be unclear. **Safe failure:** Return Escalation route unconfirmed and have the emergency-planning owner verify the primary and fallback paths.

ANSWER BOUNDARY

Do not publish private contact details broadly, infer absence reasons, guarantee a response, appoint a duty holder or contact emergency services without authorised activation.

WHY THIS IS USEFUL

Correct the call tree, verify the duty route and brief responders before stale or unreachable contacts delay escalation.

Emergency preparedness and welfare arrangements

HEALTH AND SAFETY OFFICERS QUESTION

HS-065

During a disruption or closure, which toilets, drinking-water, rest, accessible and hygiene provisions are actually available, and what temporary welfare gap needs owner action?

SENSORS AND TELEMETRY

Proposed water pressure/flow, leak alarms, consumable levels, powered-door/lift state and anonymous queue counts may support service availability. **Timing:** use 5-minute status and 15-minute demand summaries. **Evidence quality:** show sensor coverage, calibration, refill thresholds, clock alignment, uptime and unmonitored facilities; no person-level use or health inference. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the Facilities/welfare service owner and local building manager; supporting sources: closure notices, cleaning records, accessibility arrangements, water-quality owner and event plans. **Precedence:** a formal closure, water-quality instruction or competent accessibility assessment outrank sensor status; current alternative-provision records govern. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 5 to 15 minutes during disruption and the latest service notice. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and recommendation. **Decision interval:** every 15 minutes during disruption and before events. **Minimum evidence:** current facility status, planned occupancy/activity and approved minimum/alternative arrangements. **Method:** welfare-facility availability, accessibility and demand screening; identify unavailable or overloaded provision, route dependencies and gaps for accessible use without inferring individual needs. **Uncertainty:** Service telemetry cannot confirm cleanliness, physical accessibility, drinking-water quality or actual queue conditions at every facility. **Safe failure:** Return Welfare availability not verified and require inspection or a temporary alternative before relying on the provision.

ANSWER BOUNDARY

Do not infer personal hygiene, disability or health, certify water quality or accessibility, disclose individual use, or reopen a closed facility.

WHY THIS IS USEFUL

Provide temporary welfare alternatives, change occupancy or opening arrangements and prioritise repair or replenishment.

HEALTH AND SAFETY OFFICERS QUESTION

HS-066

Do official forecasts or qualified local measurements meet an owner-approved trigger for heat, cold, air quality, flooding, power or water contingency action?

SENSORS AND TELEMETRY

Proposed evidence may include official weather or flood alerts, local indoor and outdoor temperature and humidity, approved particulate sensors, water-ingress alarms and utility-service status. **Timing:** use official issue times, 5-minute local measurements and 15-minute persistence summaries. **Evidence quality:** show source authority, sensor siting and calibration, alert geography, clock alignment, coverage, uptime, forecast uncertainty and unmonitored hazards. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the contingency-plan owner, Estates or resilience lead and relevant official alert provider; supporting sources: local operating limits, vulnerable-service dependencies, closure decisions, duty rosters and welfare arrangements. **Precedence:** official warnings and the authorised contingency decision outrank local analytical estimates; current competent owner instructions govern action. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** live during a contingency and at each official alert update; expose conflicts. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: forecast, observation, comparison and recommendation. **Decision interval:** on each official alert, every 15 minutes while near a trigger and whenever a critical utility changes. **Minimum evidence:** a current plan trigger, applicable geography or mapped zone and one authoritative alert or commissioned local measure. **Method:** owner-defined contingency-trigger evaluation; compare values and persistence with the approved trigger, retain forecast ranges and separate official, observed and estimated evidence. **Uncertainty:** Forecast uncertainty, local sensor representativeness, alert latency and a changed trigger definition can alter the contingency result. **Safe failure:** Return Trigger status uncertain and refer to the contingency owner using the official warning and a conservative local check.

ANSWER BOUNDARY

A forecast or screening measurement must not be presented as certainty, diagnosis or authority to close, evacuate or continue operation. Do not issue a weather, health or exposure diagnosis, invent a trigger, predict individual harm, order closure or replace official warnings and authorised incident or resilience command.

WHY THIS IS USEFUL

Trigger the approved contingency review, inspect local conditions and brief the responsible decision owner with forecast uncertainty visible.

Emergency preparedness and welfare arrangements

HEALTH AND SAFETY OFFICERS QUESTION

HS-067 Which emergency-response assets are missing, inaccessible, discharged, expired or unverified for the hazards and zones they are meant to cover?

SENSORS AND TELEMETRY

Proposed cabinet contacts, tamper/seal status, inventory tags, pressure/self-test outputs and route-state sensors may cover selected equipment. **Timing:** use daily health and access-change events. **Evidence quality:** show exact inventory coverage, test scope, expiry source, calibration, clock alignment, uptime and manual-inspection gaps; a closed cabinet does not prove completeness. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the emergency-equipment owner and local area manager; supporting sources: asset register, inspection/test records, service certificates, risk assessments and route/closure records. **Precedence:** a competent inspection and current service record outrank sensor or tag presence; the task/location risk assessment defines required equipment. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** daily opening, before high-risk work and immediately after use or fault. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and comparison. **Decision interval:** daily, pre-task and after an incident. **Minimum evidence:** required equipment list, mapped location, current inspection/test and access status. **Method:** equipment inventory, inspection and accessibility readiness checking; identify missing, used, expired, obstructed and unverified items, and require physical confirmation where sensing is partial. **Uncertainty:** A cabinet opening or inventory tag cannot prove equipment condition, charge, contents, expiry, accessibility or suitability for the hazard. **Safe failure:** Return Emergency-asset readiness unknown and require competent inspection or replenishment before dependent work.

ANSWER BOUNDARY

Do not certify equipment operation, choose or use response equipment, infer an incident from an opened cabinet, or authorise work because equipment appears present.

WHY THIS IS USEFUL

Replenish, service or relocate response equipment and hold dependent work until competent inspection confirms readiness.

HEALTH AND SAFETY OFFICERS QUESTION

HS-068 For the identified hazard, is the building's shelter-in-place or controlled-movement arrangement operational for the intended zones?

SENSORS AND TELEMETRY

Proposed evidence may include official hazard alerts, outdoor and indoor air indicators relevant to the plan, ventilation and damper state, door control, communications heartbeat, power and water status. **Timing:** use live hazard issue times, 1-minute service states and 5-minute zone summaries. **Evidence quality:** show alert geography, sensor suitability, commissioning, mapping, calibration, clock alignment, latency, uptime and rooms without coverage; indoor readings do not prove protection. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: authorised incident command, the shelter or movement plan owner and technical service owners; supporting sources: current zone instructions, accessible communications, occupancy assumptions, staffing cover, impairment logs and exercise findings. **Precedence:** current incident command and competent service-owner decisions outrank telemetry or a generic plan. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** live during the identified hazard and at each command or service change; expose conflicts. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, observation and comparison. **Decision interval:** at activation, every 5 minutes during the event and whenever command or a required service changes. **Minimum evidence:** the identified hazard, current command decision, applicable zones and owner-defined dependency checklist. **Method:** plan-to-zone dependency readiness check; verify each required route, service, communication and welfare dependency, listing failed, unverified and inspection-only items. **Uncertainty:** The hazard can evolve, plan scope may not match the zone and power, communication, ventilation or door dependencies can fail together. **Safe failure:** Return Shelter or movement readiness unresolved and give the bounded dependency list to authorised incident command without issuing an instruction.

ANSWER BOUNDARY

Do not declare the building safe, choose shelter instead of evacuation, issue movement commands, infer exposure or override emergency services or authorised incident command.

WHY THIS IS USEFUL

Give incident command a bounded dependency picture, direct urgent checks, repair failed communications and identify zones needing an alternative arrangement.

Emergency preparedness and welfare arrangements

HEALTH AND SAFETY OFFICERS QUESTION

HS-069

Which emergency and welfare arrangements have an unmitigated single point of failure across power, network, water, access, equipment or staffing dependencies?

SENSORS AND TELEMETRY

Proposed power meters, UPS or generator health, network-service monitoring, water pressure, BMS status, door-release state and communications heartbeat may support dependency checks. **Timing:** use 1-minute states and 15-minute summaries for mapped critical services. **Evidence quality:** show dependency-map version, test mode, calibration, clock alignment, coverage, uptime and common-cause blind spots. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the business-continuity owner and each technical service owner; supporting sources: asset-dependency diagrams, resilience tests, maintenance records, emergency plan and supplier arrangements. **Precedence:** competent engineering and continuity assessments outrank inferred graph links; verified test results and current design records govern. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** monthly for planning and within 5 minutes during a disruption. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and estimate. **Decision interval:** quarterly, after change and live during disruption. **Minimum evidence:** current critical-service list, dependency map and verified component states. **Method:** dependency-graph, common-cause and single-failure scenario analysis; trace upstream utilities, communications and access, test one-failure and common-cause scenarios, and label untested assumptions. **Uncertainty:** The dependency map may omit common-cause failure, backup duration, capacity, manual fallback or staffing needed to operate the fallback. **Safe failure:** Return Resilience not demonstrated and require owner-led scenario testing and a verified manual fallback.

ANSWER BOUNDARY

Do not guarantee resilience, operate backup systems, authorise continued occupation, infer design capacity from normal operation or replace engineering assessment.

WHY THIS IS USEFUL

Prioritise redundancy, manual fallback, tests, spares and response agreements for dependencies with greatest consequence and no mitigation.

HEALTH AND SAFETY OFFICERS QUESTION

HS-070

What did the latest exercise demonstrate about detection, command, communication, movement, welfare and recovery, and which conclusions remain unsupported?

SENSORS AND TELEMETRY

Proposed alarm, door, communications and anonymous zone-count event logs may support exercise timing without tracking people. **Timing:** use the declared exercise window and aggregated movement milestones. **Evidence quality:** show clock alignment, exercise/test mode, route coverage, counter uncertainty, uptime and excluded real events; no individual performance or emotion analysis. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the exercise director and emergency-planning owner; supporting sources: observer notes, exercise plan, participant feedback, system logs, Security and Estates records. **Precedence:** the approved debrief and competent observer findings outrank telemetry; conflicting observations remain visible for review. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** immediately after the exercise and at formal debrief. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation, comparison and recommendation. **Decision interval:** post-exercise and at action review. **Minimum evidence:** declared objectives, scenario timeline, observer evidence and relevant system logs. **Method:** exercise timeline, objective-coverage and evidence-gap analysis; align detection, notification, movement and recovery milestones, compare to objectives and identify unobserved or ambiguous periods. **Uncertainty:** Exercises are artificial, objectives may not cover every condition and participant, system or sensor behaviour may differ in a real event. **Safe failure:** Record the conclusion as Not tested or Unsupported and plan a targeted exercise instead of claiming readiness.

ANSWER BOUNDARY

Exercise timing and counts must not be used to score named people or certify emergency readiness. Do not rank or identify participant performance, infer panic, certify readiness, treat an exercise as proof of real-event outcomes or alter plans automatically.

WHY THIS IS USEFUL

Update the emergency plan, exercise design, communications or welfare arrangements using only findings supported by the exercise evidence.

Audits, trends, assurance boundaries and improvement planning

HEALTH AND SAFETY OFFICERS QUESTION

HS-071 For the scheduled assurance activity, which required evidence is current, permissioned, traceable and reproducible, and which evidence is missing?

SENSORS AND TELEMETRY

Proposed source-health and export metadata can show coverage, retrieval time, checksums, version and missing records. **Timing:** use a point-in-time audit snapshot. **Evidence quality:** show commissioning, permission, retention, clock alignment and unavailable systems; telemetry summaries must link back to controlled source records and quality statements. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the audit owner and each evidence source owner; supporting sources: document control, risk, incident, training, Estates, contractor and emergency records. **Precedence:** the approved audit scope and controlled records outrank dashboards; source owners determine authenticity and authorised disclosure. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the audit cut-off, with later changes separately timestamped. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup and comparison. **Decision interval:** at planning, cut-off and pre-submission review. **Minimum evidence:** approved scope, evidence requirements, source-owner access and current records. **Method:** audit-evidence inventory, completeness and provenance validation; map every requirement to evidence, version, owner, time and permission; flag missing, stale, conflicting and analytically derived items. **Uncertainty:** Exports can omit records, permissions or transformations; source versions may change after the reporting cut-off. **Safe failure:** Return Evidence pack incomplete, disclose the exact gap and request the source owner without implying an assurance conclusion.

ANSWER BOUNDARY

Do not fabricate evidence, imply an audit pass, alter source records, disclose restricted information or present an analytical summary as formal assurance.

WHY THIS IS USEFUL

Request missing evidence early, preserve provenance and provide the assessor with a navigable pack that clearly exposes exclusions and gaps.

HEALTH AND SAFETY OFFICERS QUESTION

HS-072 After normalising for activity volume and reporting coverage, which hazard, location or asset themes show a sustained adverse trend?

SENSORS AND TELEMETRY

Proposed aggregate exposure proxies may include occupancy-hours, booking-hours, equipment run-hours and permit-hours at zone or asset-class level. **Timing:** use monthly aggregates with small-number suppression. **Evidence quality:** show denominator coverage, taxonomy changes, calibration, clock alignment, missing periods and changed sensing; no individual or team score. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the incident/risk data owner and relevant activity/asset owners; supporting sources: inspection results, work orders, bookings, permits and asset register. **Precedence:** validated incident coding and owner-approved denominators outrank raw counts; competent investigation findings outrank trend association. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** monthly and quarterly reporting periods. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: calculation, estimate and comparison. **Decision interval:** monthly, quarterly and after a major operational change. **Minimum evidence:** stable definitions, at least three periods, credible numerator and denominator, and known coverage. **Method:** rate-trend, change-point and confidence-interval analysis; calculate rates, test sensitivity to reporting and activity changes, identify sustained versus random variation and show uncertainty. **Uncertainty:** Exposure denominators, reporting culture, classification, seasonality, small counts and changing coverage can create or hide a trend. **Safe failure:** Return Adverse trend not established, show confidence and coverage, and request focused investigation where concern remains.

ANSWER BOUNDARY

Trend estimates must not rank individuals or teams, prove cause or imply that under-reporting is absence of risk. Do not infer causation, blame or underperformance, create personal risk profiles, compare groups with incompatible reporting cultures or claim no risk from low counts.

WHY THIS IS USEFUL

Direct investigation and prevention resources to credible adverse patterns, not raw counts distorted by activity or reporting coverage.

Audits, trends, assurance boundaries and improvement planning

HEALTH AND SAFETY OFFICERS QUESTION

HS-073

Which inspection, maintenance, incident and corrective-action backlogs create the largest present assurance gap, and why?

SENSORS AND TELEMETRY

Proposed asset-health and action-status feeds may corroborate whether controls remain unavailable. **Timing:** use daily snapshots and verified fault events. **Evidence quality:** show link quality, latency, clock alignment, uptime, duplicate tickets and unmonitored controls; an open ticket does not always mean control failure, and a closed ticket does not prove effectiveness. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the risk/action, inspection and maintenance programme owners; supporting sources: CMMS, project plans, budget/procurement and current asset/room owners. **Precedence:** current risk consequence and competent owner priority outrank ticket age alone; formal impairment and overdue inspection status remain authoritative. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** daily for critical backlogs and weekly for portfolio review. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: calculation, estimate and recommendation. **Decision interval:** daily and weekly governance review. **Minimum evidence:** current backlog items linked to risk/control, consequence, due date, owner and status evidence. **Method:** risk-weighted backlog, dependency and ageing analysis; rank transparently by consequence, exposure opportunity, control dependency, age and evidence uncertainty; show sensitivity to missing data. **Uncertainty:** Backlog completeness, consequence ratings, dependencies, deferrals and closure states may differ across source systems. **Safe failure:** Return Priority uncertain, escalate high-consequence unknowns and do not automatically displace mandatory or owner-set commitments.

ANSWER BOUNDARY

Do not determine legal priority, close or reassign work, infer negligence, promise completion or substitute a score for local risk judgement.

WHY THIS IS USEFUL

Allocate inspection, maintenance and action capacity to backlogs with the largest credible consequence and weakest current control.

HEALTH AND SAFETY OFFICERS QUESTION

HS-074

Which dashboard claims are unreliable because of unmapped sensors, blind spots, clock drift, calibration lapse, downtime, aggregation or missing permission?

SENSORS AND TELEMETRY

Proposed source-health telemetry should expose sensor and gateway heartbeat, mapped asset or zone, calibration due date, clock offset, latency, retained gaps, aggregation configuration and permission failures. **Timing:** evaluate live health every 5 minutes and publish a daily quality summary by evidence route. **Evidence quality:** a healthy device does not prove correct placement, representativeness or semantic mapping; independent inspection and owner confirmation remain necessary. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the telemetry platform owner and each physical sensor/asset owner; supporting sources: commissioning certificates, calibration records, mapping register, change tickets and data-quality rules. **Precedence:** current commissioning and calibration status outrank observed data volume; source-owner fault declarations outrank automated health scores. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 5 minutes for live health and the current commissioning/calibration record. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: observation, calculation and comparison. **Decision interval:** with every answer, daily dashboard review and after system change. **Minimum evidence:** the claim's required sensors plus current health, mapping, time and calibration evidence. **Method:** data-quality rule evaluation and claim-level reliability tracing; propagate source quality into the answer, identify unsupported zones/intervals and return unknown when minimum quality fails. **Uncertainty:** A heartbeat cannot prove correct location, mapping, calibration or representativeness, and aggregation or permission gaps can hide failure. **Safe failure:** Withdraw or caveat the claim and recommission the evidence route before it supports another decision.

ANSWER BOUNDARY

A healthy device heartbeat must not be treated as proof of correct placement, calibration, mapping or representativeness. Do not infer absence of hazard from absence of data, hide downtime, extrapolate across unmapped rooms, certify sensor accuracy or alter calibration records.

WHY THIS IS USEFUL

Withdraw or caveat unreliable claims, repair mapping or source health and re-commission the evidence route before reuse.

Audits, trends, assurance boundaries and improvement planning

HEALTH AND SAFETY OFFICERS QUESTION

HS-075

Across comparable locations and periods, is a strategic control programme associated with sustained improvement, and what alternative explanations remain?

SENSORS AND TELEMETRY

Proposed before/after indicators should be selected in advance and measured with stable sensors, placement, aggregation and privacy settings. **Timing:** use daily or weekly values over comparable periods. **Evidence quality:** show calibration, clock alignment, uptime, coverage changes, seasonality, activity volume and competing interventions; no personal outcome tracking. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the improvement owner and risk owner; supporting sources: change-control record, baseline assessment, incident/inspection data, commissioning and affected activity owners. **Precedence:** the predefined evaluation plan and competent review outrank post-hoc favourable metrics; adverse findings and changed operations remain visible. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** after the agreed minimum observation period and at scheduled review. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: calculation, comparison and estimate. **Decision interval:** at the pre-agreed programme review points. **Minimum evidence:** a documented theory of change, stable indicator definition, comparable baseline and follow-up periods, exposure denominator, implementation dates, source-quality record and relevant concurrent changes. **Method:** use a pre-specified interrupted time series, matched comparison or stratified before/after analysis; test sensitivity to season, reporting and activity volume. **Uncertainty:** Confidence intervals, power, residual confounding, implementation variation, concurrent changes and alternative explanations limit attribution. **Safe failure:** Report Association not established and continue or redesign the evaluation rather than claiming that the programme caused improvement.

ANSWER BOUNDARY

Do not infer causation from a simple before/after change, hide adverse patterns through aggregation, or declare the programme effective without comparable evidence. No safety or compliance certification or continuation authorisation.

WHY THIS IS USEFUL

Decide whether to continue evaluation, adapt implementation, target further investigation or commission a stronger comparison without overstating impact.

HEALTH AND SAFETY OFFICERS QUESTION

HS-076

Which proposed actions should enter the next improvement plan when consequence, exposure opportunity, commitments, feasibility, dependencies and evidence strength are considered together?

SENSORS AND TELEMETRY

Proposed telemetry may quantify control availability, exposure opportunities, repeat faults and evidence gaps at aggregated zone/asset level. **Timing:** use monthly or quarterly summaries. **Evidence quality:** show denominator, coverage, calibration, clock alignment, uptime and uncertainty; sensing should support, not replace consultation, engineering judgement or formal commitments. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the health and safety governance/risk owners and accountable budget owners; supporting sources: risk register, incident lessons, audit findings, consultation, asset plans, commitments and feasibility studies. **Precedence:** mandatory or formally committed actions and competent risk judgement outrank analytical ranking; approved strategy and budget decisions remain authoritative. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** annual/quarterly planning and after major incidents or changes. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: calculation and recommendation. **Decision interval:** quarterly and annual planning cycles. **Minimum evidence:** candidate actions, criteria definitions, owner ratings, cost/feasibility ranges and evidence quality. **Method:** transparent multi-criteria decision analysis with sensitivity testing; show weighted and unweighted rankings, dependencies, uncertainty and distributional effects; retain owner rationale and dissent. **Uncertainty:** Weights, cost and feasibility ranges, mandatory obligations, dependencies and distributional effects can make a ranking unstable. **Safe failure:** Show sensitivity and unresolved criteria, then route the choice to governance; do not accept risk or approve budget.

ANSWER BOUNDARY

A weighted ranking is advisory and must not accept risk, displace mandatory commitments or replace consultation and competent judgement. Do not determine legal obligation, approve budget, accept residual risk, hide consultation needs, or present a ranking as an authorisation.

WHY THIS IS USEFUL

Build a defensible improvement plan, sequence dependencies, assign owners and direct resources to actions with the strongest risk-reduction case.

Audits, trends, assurance boundaries and improvement planning

HEALTH AND SAFETY OFFICERS QUESTION

HS-077 Where do authoritative systems conflict about owner, asset, zone, closure, permit, action or completion status, and who must resolve each conflict?

SENSORS AND TELEMETRY

Proposed observed states may help expose a conflict but cannot settle governance. **Timing:** use event timestamps and daily reconciliation. **Evidence quality:** show mapping confidence, clock alignment, latency, uptime and sensor-interpretation limits; keep each source and its provenance separate rather than overwriting with a blended value. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the owners of every conflicting source and the health and safety service owner; supporting sources: risk, space, access, CMMS, permit, incident and project systems. **Precedence:** the published source-precedence rule applies by field; where equal or undefined, no automated winner is selected and owners must resolve. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** 5 minutes for live operational states and one day for administrative records. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: comparison and recommendation. **Decision interval:** with every query and daily reconciliation. **Minimum evidence:** two permissioned sources referring to the same asset/location/control and their effective/retrieval times. **Method:** field-level conflict detection, precedence application and unresolved-issue register; compare identifiers, versions, states and time validity, apply approved precedence and retain the losing value and reason. **Uncertainty:** Identifiers, effective times, replication lag, permissions and an undefined field-level precedence rule can prevent resolution. **Safe failure:** Return Source conflict unresolved, suppress the derived decision and route the exact field conflict to its authoritative owner.

ANSWER BOUNDARY

Do not silently choose a convenient source, overwrite records, infer owner intent, disclose restricted conflicts broadly or certify the resolved state.

WHY THIS IS USEFUL

Send each conflict to the authoritative owner, prevent contradictory instructions and correct master data at source.

HEALTH AND SAFETY OFFICERS QUESTION

HS-078 For each catalogue question and user role, are purpose, access, minimisation, retention, small-group protection and disclosure controls adequate?

SENSORS AND TELEMETRY

Proposed governance telemetry may report role-based query counts, denied fields, small-group suppression, retention jobs and source-permission health without storing question content, identities or movement histories. **Timing:** use daily service totals and quarterly role-by-purpose review, suppressing small groups. **Evidence quality:** show configuration version, audit coverage, missing connectors, clock alignment and access-log retention; system logs cannot establish lawful purpose or fairness. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the information-governance/data-protection owner and each data source owner; supporting sources: approved purpose, privacy assessment, access-control policy, retention schedule and health and safety governance. **Precedence:** lawful or approved purpose, least-privilege decision and source-owner permission outrank analytical convenience; person consent or restricted records remain controlling. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** at design, permission change, new question and periodic review. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** before access, on change and quarterly review. **Minimum evidence:** a defined question, user role, decision purpose, required fields, aggregation and retention need. **Method:** purpose-role-data policy mapping and least-privilege validation; test necessity, proportionality, minimisation, small-group risk, onward disclosure and deletion; deny or aggregate where controls fail. **Uncertainty:** Purpose can drift, combined data may become linkable, small groups may be re-identifiable and retention or role controls may fail. **Safe failure:** Deny, minimise or aggregate the evidence route until the information-governance and source owners approve the control set.

ANSWER BOUNDARY

Service logs must not be repurposed to inspect question content, reconstruct movement or create individual risk or medical records. Do not expand purpose, infer health or identity, combine datasets to track people, reveal restricted records, bypass permissions or retain data longer than authorised.

WHY THIS IS USEFUL

Approve, redesign or deny the evidence route, tighten permissions and retention, and protect small groups before operational use.

Audits, trends, assurance boundaries and improvement planning

HEALTH AND SAFETY OFFICERS QUESTION

HS-079

Does the governance report clearly separate controlled lookup, direct observation, calculation, estimate, forecast, comparison, recommendation, unknown and stale evidence?

SENSORS AND TELEMETRY

Proposed report metrics should carry source, numerator/denominator, aggregation, coverage, calibration, clock alignment, uptime, gaps and suppression status. **Timing:** use frozen reporting snapshots. **Evidence quality:** show changes in definitions and instrumentation; no metric should imply individual health, blame, productivity or complete building coverage. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the health and safety reporting owner and every metric source owner; supporting sources: approved metric dictionary, risk/incident/action systems, Estates and governance meeting scope. **Precedence:** controlled source records and metric definitions outrank visual summaries; owner corrections after cut-off are shown as dated amendments. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** the formal reporting cut-off and retrieval time. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, calculation and comparison. **Decision interval:** before each governance report is released. **Minimum evidence:** metric specification, numerator and denominator, source lineage, transformation code or rule, coverage, suppression, owner, period and evidence class. **Method:** tag every claim as controlled lookup, direct observation, calculation, estimate, forecast, comparison, recommendation, unknown or stale; verify that caveats follow the claim. **Uncertainty:** Missingness, measurement error, aggregation, denominator instability, changing definitions and unresolved source conflicts can alter a claim class. **Safe failure:** Withdraw or label the claim Unknown until its evidence class, transformation and lineage are reproducible.

ANSWER BOUNDARY

A polished report must not hide uncertainty or turn a recommendation into assurance, certification or authorisation. Do not claim compliance or assurance, hide adverse uncertainty, compare incompatible periods, infer personal outcomes or promise that reported trends will continue.

WHY THIS IS USEFUL

Let governance readers distinguish facts from estimates and recommendations, challenge weak claims and make decisions against visible uncertainty.

HEALTH AND SAFETY OFFICERS QUESTION

HS-080

Which health and safety decisions must remain outside the service because they require inspection, consultation, occupational-health, emergency, legal or other competent judgement?

SENSORS AND TELEMETRY

Proposed evidence-coverage maps should list, for each question, available sensors and integrations, mapped assets or zones, timing, aggregation, calibration, clock status, uptime, permission and gaps. **Timing:** use live source health plus a quarterly capability snapshot. **Evidence quality:** explicitly mark uncommissioned, unmapped, stale, permission-denied and proxy-only routes; never substitute a weak proxy for missing evidence. **Traceability:** record modality/feed ID, exact installation point and mapped asset or zone, spatial and temporal coverage, sample and aggregation interval, calibration or functional-test state, clock offset, uptime, gaps, drift or resets, permission and provenance. Unmapped, uncommissioned, stale, permission-denied or unhealthy evidence is unavailable.

AUTHORITATIVE SUPPORTING DATA

Current owners and sources: the health and safety governance owner and each competent decision owner; supporting sources: legal, occupational-health, emergency, inspection, consultation and technical competence boundaries, service risk assessment and approved use cases. **Precedence:** applicable law, authorised emergency command, competent professional judgement and current organisational delegations outrank catalogue coverage or analytical capability. **Metadata:** record ID/version, owner, permission, effective time, observed or event time where relevant, retrieved time and lineage. **Staleness:** every governance, legal, professional or service-capability change; expose conflicts. **Conflicts:** retain competing states and route resolution to the relevant current owner and affected source owner. Retrieval recency, analytics and telemetry cannot override a controlled source.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. **Decision interval:** before enabling a question, quarterly and after any material system or governance change. **Minimum evidence:** a named decision, lawful purpose, accountable owner, evidence route and explicit boundary. **Method:** capability and prohibited-decision boundary mapping; classify answerable, partially answerable, inspection-only, professional-judgement, emergency-command and prohibited cases; test safe failure and permission. **Uncertainty:** Law, delegation, professional boundaries and source permissions can change, while technical capability can be mistaken for authority. **Safe failure:** Refuse the unsupported decision and route it to the named inspection, consultation, occupational-health, emergency, legal or competent process.

ANSWER BOUNDARY

Do not expand capability from technical availability, present the catalogue as legal advice, infer health, create medical or individual risk records, certify safety or compliance, accept risk, authorise work, command an emergency response or control safety-critical assets. Do not use the May 2021 floor plan as current operational truth.

WHY THIS IS USEFUL

Keep the service inside an approved decision-support role, route excluded decisions correctly and prioritise evidence improvements without creating unsafe automation.

Catalogue summary: 80 bounded decision questions

The catalogue supports officer decisions without turning observations or analytics into safety, compliance or medical conclusions.

QUESTION COVERAGE

- 01

Policies, responsibilities and risk registers
Confirm governing documents, accountable owners, review triggers, control dependencies and decision-ready escalation.

HS-001 - HS-010
- 02

Hazards, inspections and workplace conditions
Use reports, inspections and correctly scoped measurements to target competent checks and immediate controls.

HS-011 - HS-020
- 03

Occupancy, activities and environmental evidence
Relate anonymous space use and declared activities to current limits without tracking people or inferring health.

HS-021 - HS-030
- 04

Incidents, near misses and corrective actions
Support triage, evidence preservation, pattern analysis and action closure without blame or diagnosis.

HS-031 - HS-040
- 05

Contractors, permits and high-risk work
Check work interfaces, permits, isolations, readiness and handback without authorising work or controlling plant.

HS-041 - HS-050
- 06

Training, competence and communications
Find record, briefing and communication gaps while separating completion records from practical competence.

HS-051 - HS-060
- 07

Emergency preparedness and welfare arrangements
Test evidence for emergency systems, routes, role cover, welfare and exercises within current command arrangements.

HS-061 - HS-070
- 08

Audits, trends, assurance boundaries and improvement planning
Make assurance evidence reproducible, expose blind spots and prioritise proportionate improvement.

HS-071 - HS-080

EVIDENCE CLASSES

Controlled lookup; direct observation; calculation; estimate; forecast; comparison; recommendation; Unknown; stale.

SAFE OUTCOMES

Answer with bounded evidence; request inspection; request owner confirmation; preserve evidence; hold for evidence; escalate to competent judgement.

OFFICER ACTIONS

Prioritise inspection, repair, consultation, briefing, temporary controls, governance escalation, source correction or improvement planning.

NEVER DELEGATED

Diagnosis, legal or occupational-health judgement, safety or compliance certification, risk acceptance, permit issue, work authorisation, emergency command or asset control.

Using the catalogue: evidence and permissions rules

Apply these rules before any question is answered, shared or used to support action.

01

Proposed by default

A named sensor, integration or analytic is not evidence of deployment. Verify installation, mapping, commissioning, permission and current health first.

02

Authoritative owner first

Controlled records and competent owner decisions take precedence. Keep conflicts visible and route them to the source owner.

03

Historical plan only

Use the May 2021 floor plan only for historical geometry. Never use it to claim current use, access, services, hazards, limits or closures.

04

Exact evidence envelope

Expose modality, installation point, mapped asset or zone, coverage, interval, calibration or test state, clocks, uptime, gaps, drift and provenance.

05

Three times

Keep effective time, observed or event time, and retrieved time distinct. A recently retrieved stale record remains stale.

06

Minimum evidence

Do not calculate, compare or recommend until the card-specific minimum is met. Missing evidence must remain visible.

07

Uncertainty is part of the answer

State blind spots, measurement error, aggregation loss, clock mismatch, stale sources, permissions and unresolved conflicts.

08

Safe failure

Return Unknown, hold for evidence or request competent inspection or consultation rather than filling a safety gap with an assumption.

09

Privacy minimisation

Use anonymous zone or asset evidence where justified. Do not identify, track, profile or combine data to reconstruct a person's activity.

10

No health inference

Do not infer symptoms, disability, pregnancy, exposure, injury severity or individual risk; do not create medical or individual risk records.

11

No content inspection

Do not inspect research content, code, speech, images of people or network content to answer a health and safety question.

12

Role-based permissions

Limit purpose, fields, location, time, retention and onward disclosure. Protect small groups and restricted operational records.

13

Competent judgement remains human

Inspection, consultation, occupational health, legal interpretation, emergency command and specialist assessment stay with authorised professionals.

14

No certification or authorisation

The service cannot certify safety or compliance, accept risk, issue a permit, authorise work, order evacuation or approve reopening.

15

No asset control

Do not unlock, isolate, reset, bypass, start, stop or otherwise control a safety-critical asset from an answer.

16

Provenance retained

Keep source ID, version, owner, transformation, export time and permission context so every claim can be reproduced and challenged.

BOUNDARY

When evidence is incomplete, conflicting, stale, unmapped or outside permission, the correct answer is Unknown plus the named owner, inspection, consultation or evidence needed next - not an inferred safety conclusion.